

Supplementary Information

Transfer Boundaries in Affective Audio Modelling: Zero-Shot Loss, Measured Adaptation Cost, and the Limits of the Available Evidence

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Revision record: what changed between the preprint and this submission

An earlier version of this work was prepared as a preprint under the title *Not all generalisation failures can be bought back: four boundaries in affective audio modelling*. Its organising claim was that generalisation failures divide into two kinds — budget-limited, which target-side observations close, and information-limited, which they do not — with mutually exclusive remedies. A pre-submission audit established that this claim is contradicted by the project’s own measurements. The framework is withdrawn. This section records what changed and why, so that a reader comparing the two versions can see exactly which numbers moved and which conclusions were retracted. A fuller machine-checkable record, including the recomputation scripts and their output, is in `reports/TAC_revision_audit.md` in the code repository.

Withdrawn claims

1. **The two-kind dichotomy and the mutual exclusivity of its remedies.** The few-shot learning curve that priced the recoverable gap was computed on the six *cross-domain* corpus pairs (`figures_r/fig_price.R` filtered `cross_domain`), while the preprint’s abstract and discussion attributed it to same-domain transfer and simultaneously described the cross-domain boundary as one on which no purchase could be found. Both halves were wrong, and the correction removes the one boundary at which the preprint claimed to observe both kinds of failure inside a single experiment. The figure legend had stated the correct condition throughout; the prose had not.
2. **“The barrier is one of calibration, not of sign”** (cross-synthesis boundary). The reported sign agreement pools, for each model, its own 40 training-domain excerpts with 40 from the other domain. Disaggregated, the environmental model’s spectral-smoothing effect on environmental excerpts is in the opposite direction and not significant (median -0.019 , $p = 0.073$). Independently of the data, a purely calibrational deficit cannot move a Spearman rank correlation at all, since rank correlation is invariant under any strictly monotone recalibration.
3. **The 31% ceiling fraction on the physiological boundary.** The denominator was $\sqrt{r}$ of the stimulus-level reliability of a PMEmo electrodermal measure, and a sham using presentation position alone reproduces 114–249% of that reliability (Section S3). A denominator that a position sham reproduces cannot bound a stimulus-driven predictor. The associated claim that the three routes’ confidence intervals overlap is also withdrawn: no confidence intervals were computed for the acoustic routes.
4. **$n^* = 318$ as a price for per-person calibration on electrodermal response.** Ten of the fifteen cells have a corrected between-person spread of exactly zero and therefore an infinite n^* ; 318 is the median over the five cells where it is finite, a subset the preprint did not disclose.
5. **“In the setting closest to an application the price has already been paid.”**

Contradicted by the preprint’s own preceding paragraph, and by the out-of-sample test reported in the main text, which returns a negative personalisation gain at every budget these corpora supply.

6. **All claims of prior specification.** Phrases such as “fixed in advance”, “pre-declared” and “stated as falsifiable in advance” are removed throughout. No traceable artifact predating the corresponding analyses exists: the manuscript directory was untracked in version control, the repository head predates most run records, and only 2 of 199 run records carry a declared hypothesis.
7. **“Choosing the wrong corpus is as expensive as choosing the wrong channel.”** The three per-person calibration settings differ in corpus, descriptor set and outcome axis simultaneously, so no one of the three can be credited with the spread.
8. **CLAP pretraining overlap as established fact.** No sample-level overlap check was performed. The overlap is described as possible at the corpus level.

Corrected quantities

1. **Emo-Soundscapes mixture leakage.** 613 of the 1,213 clips are mixtures of 30 source recordings occupying 161 distinct constituent sets, and the distributed metadata gives each clip its own group identifier, so grouped cross-validation does not separate them. Recomputed on the 600 original recordings alone: the within-corpus reference falls from 0.744 to 0.646, and median cross-domain transfer rises from 0.043 to 0.129. Every cross-domain number in the main text is reported on both variants and interpreted on the clean one.
2. **The pretrained-representation comparison.** On the leak-free corpus, AST’s advantage over hand-specified descriptors at the cross-domain boundary falls from a factor of 2.7 ($0.047 \rightarrow 0.125$) to a factor of 1.26 ($0.129 \rightarrow 0.162$).
3. **The CORAL sham control.** With one environmental corpus the “unrelated third corpus” lies inside the target’s own domain on three of the six ordered pairs. Split by validity, the valid sham recovers -11% and the invalid sham $+25\%$; the pooled 14% the preprint reported is entirely the invalid half. CORAL’s gain is therefore more target-specific than the preprint concluded, not less.
4. **“The return per doubling collapses after 100 labels.”** On the leak-free corpus the increments are 10, 14, 10 and 14 points — no collapse, and the largest increment is the last. The reported collapse was produced jointly by the mixture leakage and by the loss of the Soundtracks pairs at $k = 200$.
5. **Units of replication.** “36 ordered music-to-music pairs” is 6 ordered corpus pairs each evaluated with 6 algorithms.
6. **“The same 1,240 trials”** for the EEG and self-report reliability comparison. The EEG measures use 1,240 trials over 307 stimuli; the self-report scales use 1,080–1,121 trials over 285–298 stimuli.

Results added in this revision

1. The direct out-of-sample test of per-person calibration, which the repository contained and the preprint did not report: fitting each listener their own slope loses to the population slope in 15 of 15 electrodermal cells and in every music-rating cell at 18, 30 and 50 observations per person.
2. The presentation-order control on PMEmo electrodermal reliability (Section S3), which the analysis script had run and flagged but which appeared in neither manuscript file.
3. The disaggregation of the audio-editing experiment by model and by material domain.
4. Per-corpus ethics provenance (Table S13) in place of a blanket statement.

Sections S1–S6 are admission criteria: checks that had to pass before a main-text claim was allowed to be made, each reported in the same four parts — what was checked, the decision rule fixed before the check, the outcome, and the claim the outcome licenses. Sections S7–S9 then cover the representation-family extraction, the withdrawn analyses, and the Extended Data figures.

Table S1: The six admission criteria and the main-text claims they license. A criterion that failed removes a claim rather than weakening it; two of the six did fail, and the corresponding analyses are absent from the main text rather than reported as null results.

Section	What had to pass	What the outcome licenses
S1	The documented mapping from stimulus trigger code to audio file in ds002721 must reproduce, or be recoverable against a permutation null.	Failed. No acoustic-to-physiological analysis is reported in that corpus. Its trigger codes are used only as stimulus identifiers, for which their internal consistency suffices.
S2	Both electrodermal positive controls must pass, applied at the level at which the conclusion would be drawn.	Failed. The corpus and its conclusion are withdrawn. The physiological negatives that remain in the main text are the ones whose instrument was shown to work.
S3	Loudness must not act as a corpus identifier before descriptors are computed; and the physiological null must not be an artefact of having removed loudness.	Boundary 1, the domain wall (Fig. 1 and Table 1); Boundary 3, the physiological ceiling.
S4	The reliability of each target must be estimated, with a stated estimator, before any percentage of it is quoted.	The 84% within-corpus figure, and <i>only</i> that one. Two different denominators were conflated in the preprint: an annotation-reliability ceiling ($\sqrt{r}$ of a split-half estimate) and a within-corpus model reference. The survival fractions at the corpus boundary use the second. Partly failed: on the physiological target the reliability estimate is reproduced by presentation position (Section S3), so no percentage of it is quoted and the 31% bound is withdrawn.
S5	A within-song correlation between two autocorrelated series must exceed a circular-shift null before it is called a coupling.	That all four boundaries are stated at the level of the excerpt, and that moment-to-moment modulation is a separate and much smaller quantity.
S6	A cross-model attribution consensus must exceed its label-permutation reference distribution before it is treated as a stable descriptor set.	Failed. Transfer is measured at every boundary and never inferred from which descriptors a model relies on within its own corpus (Fig. 2).

S1 Admission criterion: the trigger-to-audio mapping in ds002721

Two errors in the event annotation of this dataset were found while preparing the analyses in the main text, and both would affect anyone reusing it. We report them so that the corrections are available, and we have written to the dataset authors. The second is the admission criterion for Boundary 3: it determines which questions this corpus can be asked.

S1.1 The documented response codes are not the response

What was checked. `events.json` documents two code families as both encoding a 1–9 rating: 833–841 (“Response 01–09”) and 901–909 (“Answer 01–09”). They agree on only **7.3%** of trials, so at most one can be the rating.

Decision rule. The question can be settled without any external reference, using the internal structure of the eight emotion scales. Ratings of pleasantness and happiness must covary positively; pleasantness and fear negatively; anger and fear positively. The encoding that reproduces the expected sign on the twelve relations is the rating; if neither did, no rating analysis would be reported.

Table S2: Internal structure of the eight emotion scales under each candidate response encoding.

Expected relation	833–841	901–909
pleasant $\leftrightarrow$ happy, positive	−0.015	+ 0.503
pleasant $\leftrightarrow$ tender, positive	−0.173	+ 0.519
pleasant $\leftrightarrow$ afraid, negative	+0.075	− 0.584
angry $\leftrightarrow$ afraid, positive	+0.065	+ 0.600
happy $\leftrightarrow$ sad, negative	+0.077	− 0.299
six relations expected positive	mean −0.063, 2/6	mean +0.406, 6/6
six relations expected negative	mean +0.005, 2/6	mean −0.382, 6/6

Outcome. 901–909 carries the ratings. 833–841 shows no coherent emotional structure. All rating analyses in the main text use 901–909.

A second, smaller point: the question codes 800–807 are re-emitted when a question is redisplayed, so a rating must be matched to the nearest question code within a tolerance rather than by exact timestamp equality. Exact matching drops roughly 45% of ratings.

S1.2 The stimulus codes do not resolve to the documented audio files

What was checked. `events.json` states that stimulus codes span 301–360 and that subtracting 300 gives the filename. The observed codes span **302–657**, exactly ten per music run, strictly interleaved with the fixation and music-onset markers. Assuming the codes are stimulus + 300 yields indices into the published stimulus set that do not correspond to what participants reported:

- Participants’ eight ratings against the published mean ratings of the mapped clips: all 64 pairings, $|\rho| \leq 0.16$.

- A spectral-centroid probe against published “energy” ratings: $\rho = +0.37$.
- The same probe against participants’ “energetic” ratings: $\rho = -0.01$.

The codes are nevertheless genuine stimulus identities: participants who received the same code agree strongly in their ratings (split-half 0.80–0.87). The numbering scheme, not the codes, is what is missing.

Decision rule. We attempted to recover the mapping by matching the eight-dimensional emotion profile of each code against that of each published clip, optimising the assignment with the Hungarian algorithm. The rule stated for the check was that the recovered assignment must beat a null formed by permuting stimulus labels within participant and regenerating the profiles; otherwise the acoustic-to-physiological link would not be examined in this corpus at all.

Outcome. The recovered assignment was not better than chance (observed mean profile correlation 0.672; null mean 0.654, 95th percentile 0.681; $p = 0.19$), and none of the recovered pairings matched the documented rule.

What it licenses. **The acoustic-to-physiological link could not be examined in this corpus, and is not examined in the main text.** What Boundary 3 reports instead is a comparison between two response channels measured on the same trials — what listeners say and what their EEG does — for which stimulus identity is sufficient and stimulus *content* is not required. Every use of these codes in the main text is as an identifier only.

A third observation, possibly related: the dataset README describes participants as hearing 40 clips, which would imply a common set. We observe 39–40 clips per participant but **307 distinct codes across the 31 participants**, with a median pairwise overlap of six clips. This is also the reason the stimulus-level reliability estimator in the main text is the one appropriate to an unbalanced design.

S2 Admission criterion: withdrawal of the electrodermal corpus

An earlier version of this work analysed an electrodermal corpus [1] and reported that single-stimulus peripheral responses carry no reproducible stimulus-level component. **That conclusion is withdrawn.** It was not withdrawn because the effect exists; it was withdrawn because we could not establish that the analysis pipeline could detect any effect at all.

S2.1 How it was found

The reported conclusion conflicts with a well-established literature on category-level skin-conductance responses to affective sounds. Prompted by that conflict, we tested the pipeline against effects known to be present.

Habituation was significant in the direction opposite to the established effect, and stimulus locking was absent. Both held after replacing our implementation with a published, validated toolbox [2].

Table S3: Group-level positive controls, under two implementations. The two share their input recordings and their event timing and differ only in the decomposition of the signal, so agreement between them bounds implementation error and is not independent corroboration of the controls themselves.

Positive control	Expected	Hand-written	NeuroKit2
Habituation across trials	$\rho < 0$	+0.113 , $p = 0.002$	+0.101 , $p = 0.015$
Post-stimulus > random window	significant	$p = 0.267$	$p = 0.357$

S2.2 Five candidate explanations, all excluded

Table S4: Candidate explanations for the control failures, and their tests.

Hypothesis	Test	Outcome
Timestamp semantics (re-sponse time, not onset)	event-related average swept over ± 16 s	no peak at any lag
Missing phasic decomposition	0.05 Hz high-pass added	reliability $-0.085 \rightarrow +0.110$; controls still fail
Ill-conditioned filter at 1 kHz	decimate to 20 Hz before filtering	8.3×10^{-7} vs 7.8×10^{-7} , no change
Implementation defect	replace with NeuroKit2	same failures
Unit scaling	rescale by 10^4 and 10^5	66 responses detected at all three scalings

The second row is itself a correction. The original extraction linearly detrended within a 6 s window, while the dominant variance in this signal is drift on the scale of minutes; trial-level responses were buried rather than absent. After adding the high-pass, stimulus-level reliability rose from -0.085 to $+0.110$ and the category-level ordering became consistent with the literature. **The strong form of the original claim — that no reproducible stimulus-level component exists — is therefore withdrawn twice over**, once for the extraction defect and once for the control failure below. What can be said is that per-stimulus reliability is low (0.110) against 0.750 for the same participants’ ratings of the same trials.

S2.3 The group-level verdict was itself tested

Decision rule. A group mean in the wrong direction is compatible with a usable subset: if some recordings failed technically, they could displace the mean while the remainder stayed valid. This dataset’s own publication documents extensive equipment problems, so the possibility was concrete. We therefore applied both controls *per participant*, with each participant’s own permutation null, and fixed the rule that the corpus would be retained only if a subset of participants passed both.

Table S5: Per-participant positive controls (94 analysable participants, 500 permutations each).

Per-participant control	Passing	Chance expectation
Habituation ($\rho < 0$ and $p < 0.05$)	8 / 94	4.7
Stimulus locking ($p < 0.05$)	4 / 94	4.7
Both	0 / 94	0.24

Outcome. Decisively, habituation was in the expected direction for only **35%** of participants, where a working measurement would put it well above 50%. The wrong direction is therefore the majority behaviour, not a minority artefact displacing a mean. No usable subset exists.

What it licenses. The corpus is excluded from every substantive analysis, and the main text reports the exclusion rather than the negative result it would otherwise have produced. This is what makes the remaining physiological results at Boundary 3 readable: the EEG corpus that carries them passes four dataset-level controls and an individual-level control before any reliability is estimated. A negative result from an instrument that has not been shown to work is not a negative result.

S2.4 What we take from it

Reliability testing and positive controls answer different questions and do not substitute for one another. Testing whether a target variable carries reproducible signal says nothing about whether the instrument works. We had performed the first and omitted the second, and the omission produced a publishable-looking negative result that we could not defend.

S3 Admission criterion: loudness normalisation

Loudness enters this work twice, in opposite roles, and both had to be settled before a main claim could be made. For the domain wall, absolute level is a corpus *identifier* that must be removed or the wall is measured against an inflated baseline. For the physiological ceiling, absolute level is the most plausible *true* driver of an autonomic response, so having removed it is the most obvious mundane explanation for a null and must be excluded.

S3.1 Loudness as a corpus fingerprint

What was checked. All audio is loudness-normalised to -23 LUFS (ITU-R BS.1770, integrated, gain scaling only, no limiting) before any descriptor is computed. The check is whether this changes cross-corpus transfer, using DEAM $\rightarrow$ PMEmo with five classifiers, and whether it changes which descriptors the models use.

Decision rule. The rule was written in terms of the loss in transfer performance that normalisation would cause, with three branches: a small loss would mean loudness carries genuine shared signal; a large loss would mean the models had been learning loudness alone and the descriptor work could not proceed; an intermediate loss would require a per-corpus decision.

Outcome. Transfer performance *rose*, which is a fourth case the rule did not cover. We record that rather than reclassify it after the fact.

The mechanism is specific and checkable: DEAM is a mixed library and PMEmo consists of pop choruses mastered at high level, so the two corpora differ in loudness distribution before they differ in anything else. A model trained on the first learns that loud means aroused and

Table S6: Loudness normalisation, DEAM $\rightarrow$ PMEmo. The first three rows are out-of-corpus quantities and all improve; the fourth is a within-corpus sanity check and falls slightly. The last row gives the mean rank, across models, of the strongest absolute-level descriptor among the 122.

	Raw	−23 LUFS	Δ
Cross-corpus AUC (random forest)	0.9180	0.9304	+0.0124
Cross-corpus recall (random forest)	0.4138	0.5172	+0.1034
Test excerpts inside the applicability domain	78.0%	86.7%	+8.7 pp
Within-corpus cross-validated AUC	0.9378	0.9282	−0.0096
Mean attribution rank of <code>rms_p90</code> (of 122)	39.0	93.0	+54.0

mis-calibrates on the second. Normalisation removes that layer, which is why an out-of-corpus quantity improves while a within-corpus quantity does not.

What it licenses. Every transfer number in Fig. 1 and Table 1, and the wall itself. Without normalisation the same-domain transfer against which the cross-domain collapse is measured is inflated by a corpus fingerprint, and the 59-to-74 percentage-point gap between the two swaps could not be attributed to the domain boundary. It also carries an engineering consequence stated in Methods: a deployed system must normalise before scoring, or it reproduces the same fingerprint.

One semantic caution: `mfcc1_mean` remains in the leading descriptors after normalisation, but it no longer indexes absolute loudness — it indexes spectral tilt. It must not be read as a loudness rule.

S3.2 The physiological null does not depend on loudness having been removed

What was checked. Normalisation removes a 6.7-fold spread in across-track loudness, and sound level is the best-established physical driver of electrodermal response. If the acoustic descriptors fail to predict song-level electrodermal response only because we discarded level, the physiological ceiling result would be an artefact of our own preprocessing.

Decision rule. Three feature pipelines were compared on the same 717 songs, the same folds and the same scoring: our descriptors on normalised audio, our descriptors re-extracted from the original audio at its native loudness, and the 6,373-dimensional ComParE feature set distributed with the corpus, which its authors extracted from the original audio with an independent toolchain. A difference in best ρ exceeding 0.05 between pipelines would mean the null is a preprocessing artefact.

Outcome. The three routes agree.

Table S7: Song-level electrodermal prediction under three feature pipelines. The best of six algorithms is reported for each. This is the only place in this work where a claim rests on agreement between independently constructed pipelines.

Pipeline	Audio	Best ρ
Hand-specified descriptors (normalised)	−23 LUFS	0.045
Hand-specified descriptors (re-extracted)	original loudness	0.049
ComParE, 6,373 features, extracted by the corpus authors	original loudness	0.072

The spread is 0.027, below the 0.05 threshold stated for this check (for which, as everywhere in this work, we can produce no artifact predating the analysis). Loudness used alone as a single predictor reaches +0.059, which is within the same band.

What it licenses. The statement at Boundary 3 that no acoustic route reaches a usable fraction of the attainable ceiling on the physiological target. It excludes the most plausible mundane explanation for that null with one number.

A restriction on how agreement may be used elsewhere. This is the only surviving claim in this work of the form “independently constructed pipelines agree”. An earlier version used that form in several places, most of them resting on attribution rankings shared between models, which Section S6 shows is not evidence of anything. No other analysis in the main text or this document should be read as claiming convergence of independent paths.

S3.3 Failed admission criterion: presentation position reproduces the electrodermal reliability

The preceding subsection asks whether the physiological null is an artefact of loudness normalisation and finds that it is not. A second and more damaging question was also asked by the analysis code and was not reported in the preprint: whether the stimulus-level *reliability* of the PMEmo electrodermal measures — the quantity that sets any ceiling on that boundary — is a property of the stimuli at all, or of where in the session they were presented.

The check has three parts and the criterion fails all three.

Assignment of songs to positions is not random. The intraclass correlation of position across songs is 0.041 ($p = 0.005$). A corpus in which every song is heard at an arbitrary position by every participant would give zero.

A position-only sham reproduces the reliability, and in one case exceeds it. Replacing the stimulus identifier with the presentation position alone and recomputing stimulus-level reliability by the identical estimator gives 0.290, 0.283 and 0.267 for skin-conductance response rate, phasic mean and skin-conductance level slope, against 0.116, 0.237 and 0.234 with the true stimulus labels — that is, 249%, 120% and 114% of the values the real labels produce (Table S8).

Regressing position out removes most of it. After partialling presentation position out of the measures, reliability falls from 0.116 to 0.038, from 0.237 to 0.081, and from 0.234 to 0.206. For the measure that carried the preprint’s headline figure, two-thirds of the reliability is position.

What it withdraws. The preprint expressed the cross-response-channel result as a percentage of an attainable ceiling — 31% for the best route, 24.5% and 21.3% for the two acoustic ones — where the ceiling was $\sqrt{r}$ of one of these reliabilities. That framing is withdrawn in full. A denominator a position sham reproduces cannot bound a stimulus-driven predictor, and the

Table S8: Stimulus-level reliability of three PMEmo electrodermal measures under the true stimulus labels, under a position-only sham, and after partialling position out. A sham that reproduces or exceeds the quantity it is a control for is a failed admission criterion, not a small caveat.

Measure	True labels	Position-only sham	Sham as % of true	Position partialled out
Skin-conductance response rate	0.116	0.290	249%	0.038
Phasic mean	0.237	0.283	120%	0.081
Skin-conductance level slope	0.234	0.267	114%	0.206

percentages additionally selected across denominators that differ by 40% between measures, in a direction that made the *lowest*-reliability measure yield the *highest* percentage.

What survives. The raw comparison. No acoustic route to these electrodermal measures exceeds a random projection of the same descriptor space, and the listeners’ own ratings do, weakly ($\rho = 0.117$, $p = 0.004$). That statement does not require a ceiling and does not depend on the reliability estimate.

What it means for reuse of this corpus. Any analysis treating PMEmo’s song-level electrodermal measures as stimulus-driven — including any that uses their reliability as a ceiling or their intraclass correlation as evidence of a stimulus effect — needs a position control first. The confound is a property of the released corpus, not of the pipeline applied to it.

S4 Admission criterion: the annotation reliability ceiling

Every survival fraction in the main text is a ratio, and this section is its denominator. A model that reaches $\rho = 0.77$ on a target whose own reproducible component is 0.87 has almost exhausted the target; the same ρ on a target whose reproducible component is 0.99 has not. Without the denominator the four boundaries cannot be compared with one another, and the comparison is the entire argument.

What was checked. The reliability of each target, estimated before any percentage of it was quoted, with the same estimator applied to the physiological targets as to the ratings.

Estimator. Where per-rater judgements are released, reliability is estimated by splitting raters at random into two halves, computing stimulus means within each half, correlating the two sets of means and correcting to the full rater panel by the Spearman–Brown formula; the reported value is the mean over 200 random splits, and we also report the 10th and 90th percentiles across splits because a ceiling estimated from ten raters per stimulus is itself uncertain. For DEAM this uses 17,762 individual judgements from 195 raters, with a median of ten per song.

Where only stimulus means and their across-observer standard deviations are released — PMEmo — the same quantity is obtained analytically: the within-stimulus variance is the mean of the squared across-observer standard deviations, the variance of the stimulus mean is $\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2/n$, and the single-observer intraclass correlation is stepped up to the n -observer mean by the same Spearman–Brown expression. On the electrodermal targets, where per-listener

values *are* available, both estimators were computed and agree (analytic 0.146–0.350 against split-half 0.138–0.232), which is what licenses the analytic form on the ratings side.

Table S9: Reliability of every target used in this work, and the two denominators it implies. r is the reliability of the group-averaged target; $\sqrt{r}$ is the largest correlation a perfect predictor could attain against the noisy target. Brackets give the 10th and 90th percentiles over 200 rater splits for the DEAM analysis subset, and 95% confidence intervals for the two ds002721 rows, whose estimator is an intraclass correlation over trials rather than a corrected split-half and which is therefore not directly comparable with the rows above it. Corpora that release no per-rater or per-observer data admit no estimate; transfer into them is reported in the main text as a raw correlation and never as a percentage.

Target	Estimator	r	$\sqrt{r}$
DEAM arousal, analysis subset ($n = 1,233$) <i>10th–90th percentile over splits</i>	rater split-half, Spearman–Brown	0.868 [0.838, 0.889]	0.932
DEAM arousal, all songs ($n = 1,802$)	rater split-half, Spearman–Brown	0.837	0.915
DEAM valence, all songs	rater split-half, Spearman–Brown	0.792	0.890
PMemo arousal ($n = 717$)	analytic ICC, Spearman–Brown	0.935	0.967
PMemo valence ($n = 717$)	analytic ICC, Spearman–Brown	0.893	0.945
ARAUS soundscape ratings	split-half on the 4.4% of stimuli with two or more raters	0.566	0.752
IADS-sound ratings (BIRAFFE2)	listener split-half, Spearman–Brown	0.750–0.813	0.866–0.902
PMemo phasic_mean	electrodermal, analytic ICC, Spearman–Brown	0.350	0.592
PMemo scl_slope	electrodermal, analytic ICC, Spearman–Brown	0.285	0.534
PMemo electrodermal, scr_rate	analytic ICC, Spearman–Brown	0.146	0.382
ds002721 ratings, best of 8 scales <i>95% confidence interval</i>	stimulus-level ICC, unbalanced design	0.221 [0.155, 0.281]	—
ds002721 EEG, best of 156 measures <i>95% confidence interval</i>	stimulus-level ICC, unbalanced design	0.092 [0.037, 0.142]	—
Emo-Soundscapes	no per-rater data released	—	—
Soundtracks	no per-rater data released	—	—

Two denominators, and which is used where. A percentage of a ceiling is meaningless until the convention is named, because two are in use and they differ by an amount that grows as reliability falls. Dividing by r answers “what fraction of the reproducible variance in the target has been recovered”. Dividing by $\sqrt{r}$ is the classical correction for attenuation and answers “what fraction of the largest attainable correlation has been reached”; it is the more conservative of the two at low reliability, and it is the convention we adopt wherever a physiological target is involved. On DEAM the two give 89% and 83% for the same model; on an electrodermal target with $r = 0.146$ they differ by a factor of 2.6. **Main-text percentages against a rating target use ρ/r ; percentages involving a physiological target use $\rho/\sqrt{r}$, which is the convention under which the 31% bound is quoted.** Mixing the two within one comparison

— which an earlier version of this work did, taking the subjective percentage from one corpus under one convention and the physiological percentage from another corpus under the other — is a defect and is recorded as such in Section S8.

The homologous table. The comparison that Boundary 3 rests on was recomputed so that both sides come from the same corpus, the same songs, the same features, the same folds and the same reliability estimator. This is the table that supports the ceiling claim; the per-corpus values above supply the denominators used elsewhere.

Table S10: One ruler. The same 717 PMEmo songs, the same 161 descriptors, the same artist-grouped folds and the same reliability estimator, applied to a rating target and to three electrodermal targets. Achievement is $\rho/\sqrt{r}$ with a double bootstrap interval.

Target	r	$\sqrt{r}$	Acoustic ρ	Achievement, 95% CI
Arousal (rating)	0.935	0.967	0.817	84.4% [81.5, 87.0]
Valence (rating)	0.893	0.945	0.705	74.6% [70.4, 78.5]
EDA phasic_mean	0.350	0.592	0.032	5.4% [−6.8, +17.4]
EDA scl_slope	0.285	0.534	0.029	5.5% [−7.5, +18.8]
EDA scr_rate	0.146	0.382	0.014	3.7% [−15.2, +22.6]

The three physiological intervals all contain zero. The bound quoted in the main text is not from this table but from the route comparison that follows it.

A correction to how that comparison was scored. An earlier version divided every route’s correlation by a single ceiling, $\sqrt{r} = 0.592$, taken from **phasic_mean**. But the three electrodermal measures have different reliabilities and therefore different ceilings — 0.382, 0.592 and 0.534 — and a route’s best correlation is not always obtained on **phasic_mean**. The first acoustic principal component, for instance, reaches $\rho = 0.114$ against **scl_slope**, whose ceiling is 0.534, and scoring it against 0.592 understated it. This is the same mismatched-denominator error recorded in Section S8 for the annotation ceiling, in a second place. Every route is now scored against the ceiling of the measure its correlation was computed on.

Rescored that way, over all 54 route $\times$ measure cells:

Table S11: Every information source we could construct, against the attainable ceiling of the electrodermal measure it was computed on. The random projection is a control, not a source.

Information source	Measure	ρ	$\sqrt{r}$	Achievement
Listeners’ valence rating	scr_rate	0.117	0.382	30.6%
Listeners’ arousal rating	scr_rate	0.106	0.382	27.8%
Rating predicted from acoustics	scr_rate	0.094	0.382	24.5%
First acoustic principal component	scl_slope	0.114	0.534	21.3%
<i>Random projection, 95th pct (control)</i>	scr_rate	<i>0.079</i>	<i>0.382</i>	<i>20.6%</i>
Direct 161-descriptor regression	scl_slope	0.029	0.534	5.5%

Two things follow, and the second is less comfortable than the first. No information source exceeds 31% of the attainable ceiling, which is the bound the main text quotes; it is a maximum over 54 cells and is biased upward for that reason. But the two acoustic routes, at 24.5% and 21.3%, sit barely above a random projection of the same descriptor space at 20.6%, so the claim

that acoustics and ratings “stop at the same place” rests on intervals that overlap rather than on estimates that coincide, and only the human rating is clearly separated from the control.

Why this is the denominator for all four boundaries. Boundary 1 expresses the cost of a corpus swap as a fraction of within-corpus performance, and within-corpus performance is itself quoted as 89% of the annotation ceiling; without the ceiling, “the model loses a fifth” could describe either a model falling away from a nearly exhausted target or one that never approached it. Boundary 2 uses the within-corpus performance of the perturbed model as its ceiling, which inherits the same denominator. Boundary 3 is the case where the ceiling itself is low, and the whole point — that the limit is on the target and not on the predictor — exists only because the ceiling was measured on the same songs with the same estimator. Boundary 4 prices per-person calibration against a break-even that depends on how much of a person’s response is reproducible at all, and the settings whose ratings are more reliable are the settings where calibration pays soonest.

Limits of the ceiling itself. Three should be stated. The DEAM estimate rests on a median of ten raters per song, so the split-to-split spread is not negligible; we report it. The ARAUS estimate uses the 4.4% of stimuli that received two or more raters, which is a small and possibly unrepresentative slice. And no ceiling exists for Emo-Soundscapes or Soundtracks, because neither releases per-rater judgements — Emo-Soundscapes was annotated by pairwise comparison rather than by direct rating. Transfer into those corpora is therefore reported as a raw correlation throughout, and the reader should not convert it to a percentage.

S5 Admission criterion: the circular-shift null

What was checked. Whether a correlation computed *within* a song, between a frame-level descriptor series and a continuous annotation series, may be interpreted at all. Both series are strongly autocorrelated: the lag-one autocorrelation of the continuous arousal annotation averages 0.931 and that of the descriptor series 0.755. Two independent series with that much internal structure produce large correlations by construction, and a p -value computed as though successive frames were independent observations is not merely optimistic but uninformative.

The null. The descriptor series is circularly shifted by a random offset (minimum ten samples, to avoid near-identity) and the correlation recomputed, 24 times per song and descriptor. A circular shift preserves each series’ own autocorrelation, its marginal distribution and its length, and destroys only the temporal alignment between them — which is exactly the hypothesis under test. This was run on 400 DEAM songs with 0.5 s continuous annotations, over 15 descriptors spanning spectral-level and temporal-variability families.

Decision rule. An earlier analysis had first-differenced both series to remove autocorrelation and found almost no coupling (strongest $|\rho| = 0.056$). Differencing is unfriendly to a series that has already been smoothed by the annotation interface, so two readings were possible and the rule was written to separate them: if the undifferenced level correlations exceed their circular-shift

null, the low coupling is an artefact of differencing; if they are indistinguishable from the null, the low coupling is a substantive finding.

Outcome: the null level is 0.25–0.30. Across the 15 descriptors the mean $|\rho|$ under the circular-shift null lies between 0.250 and 0.301, against observed level correlations of 0.267 to 0.345. An observed within-song correlation of 0.30 is, to two decimal places, what this design produces when there is nothing to find. The excess over the null is +0.02 to +0.04; 21.2% of song $\times$ descriptor combinations exceed their own null’s 95th percentile against the 5% expected by chance, with a per-descriptor range of 16–24%.

The coupling is therefore real and small: of an observed $|\rho| \approx 0.30$, about nine tenths is autocorrelation. The family ordering seen at the excerpt level reproduces here — spectral level excess 0.0366 against temporal variability 0.0237 — so the two scales do not contradict one another.

What it licenses. Two statements, one positive and one restrictive. The restrictive one is the reason this section exists: **every boundary in the main text is defined at the level of the excerpt**, and none of them is a claim about within-excerpt dynamics. Excerpt-level selection operates at $\rho \approx 0.77$ and within-song moment-to-moment coupling at an excess of about 0.04 over its null, a difference of roughly twentyfold; a claim about one is not evidence about the other. The positive statement is that the mechanistic reading we had previously assumed — that rate-of-change descriptors predict arousal *because* instantaneous change drives instantaneous arousal — is not supported at the scale it names, and is not made.

A statistic retired by the same check. A lag scan over 0–6 s returns a flat curve (mean $|\Delta\rho|$ between 0.1527 and 0.1606 across twelve lags, a variation under 5%). The “best lag” of 6.0 s is the maximum of a flat curve, not a latency estimate, and an earlier draft that cited it as evidence of a response delay was corrected.

S6 Admission criterion: the SHAP-consensus reference distribution

What was checked. Whether the number of descriptors shared by the top-20 attribution sets of three different models — used in an earlier version as the credibility threshold for a descriptor set, with a pre-set threshold of six — carries any information about whether the models found signal.

The reference distribution. The labels are permuted and the entire procedure repeated: the same three models, the same attribution settings, the same top-20 intersection. Twenty permutations were run, and the observed value was recomputed under identical settings rather than taken from an earlier run, so that observation and null are comparable. Cross-validated AUC is recorded alongside, as a check that the permutation did what it was supposed to.

Outcome. The observed consensus is 4 descriptors. The null distribution has mean 3.75, maximum 7, and places the observation at $p = 0.381$. Meanwhile the permutation plainly worked: cross-validated AUC falls from 0.936 to a null mean of 0.4964. The models learn a real signal from the intact labels, and the statistic meant to say *where* that signal lives cannot distinguish it from noise.

The mechanism is not subtle in hindsight. Which descriptors a model ranks highly is driven by properties of the descriptors themselves — variance, cardinality, how easily they support overfitting — and three models reading the same descriptor table are attracted to the same convenient columns whether or not the labels mean anything. An intersection of ranks discards effect sizes and has no null calibration of its own.

What it licenses. The design of Fig. 2 and the argument attached to it. Because a consensus count is not evidence, the four selection routes are reported as a *dissociation* rather than a convergence, and the finding that matters is that they disagree: the route that tests within-corpus significance returns temporal-variability descriptors in 2 of 12 cases (17%, below the 50% chance level) while cross-domain sign agreement, glass-box shape functions and Rashomon-set stability return them in 63%, 60% and 67% of cases. More generally, it is why every boundary in this paper is crossed and measured rather than predicted from which descriptors a model relies on inside its own corpus. The specific superseded quantity — an intersection of six descriptors reported as “first time above threshold” in the loudness experiment of Section S3 — is listed in Section S8; the transfer conclusions of that experiment are unaffected, because they never used the intersection.

S7 Extraction of the four representation families

The most direct objection to the domain wall is that 122 hand-specified descriptors are too weak and a representation learnt from a large audio corpus would cross what they cannot. Answering it requires more than one pretrained model, and in particular requires models whose pretraining data do not include the source of the environmental corpus. The four families below were chosen to span four pretraining paradigms, and three of them were extracted specifically because the fourth is contaminated — a decision taken before any of the resulting numbers existed.

Table S12: The four representation families. All embeddings are frozen; no model was fine-tuned. Downstream models, folds and scoring are identical to those used with the hand-specified descriptors. Checkpoints are named below the table by their model identifier rather than cited, because a checkpoint has no resolvable identifier of its own; the papers behind them are cited in the text.

Family	Pretraining	Paradigm	Dim.	Native rate
AST	AudioSet, 527 classes	supervised	768	16 kHz
MERT	music	self-supervised	768	24 kHz
Wav2Vec2	speech, 960 h	self-supervised	768	16 kHz
CLAP	630k audio-text pairs	contrastive	512	48 kHz

The checkpoints are MIT/ast-finetuned-audioset-10-10-0.4593, m-a-p/MERT-v1-95M, facebook/wav2vec2-base-960h and laion/clap-htsat-unfused [3], the contrastive language-audio idea being due to [3].

Each family occupies a different corner of the design. AST is supervised on a corpus that explicitly covers rain, wind, traffic and animal categories, and is therefore the family that most ought to cross the music/environmental boundary. MERT is self-supervised within the music domain: if a music-specific model also fails to cross, the failure is not a shortage of domain knowledge. Wav2Vec2 is self-supervised on speech, unrelated to either target domain, and serves as a lower bound. CLAP is the contrastive case and the one this work had used previously.

Extraction protocol. Identical across families. The middle 10 s of each excerpt is taken; the audio is loudness-normalised to -23 LUFS, the same convention used for the hand-specified descriptors (Section S3); it is resampled to each model’s own native sample rate; and the embedding is the mean over the time axis of the last hidden layer. All four corpora — DEAM, PMemo, Soundtracks and Emo-Soundscapes — were processed this way, giving sixteen corpus $\times$ family cells. CLAP was re-extracted at 10 s for this comparison even though a 30 s version already existed, so that CLAP does not differ from the other three families in excerpt duration as well as in architecture; both versions are retained.

Contamination disclosure for CLAP. LAION-Audio-630K, the pretraining set of the checkpoint used here, is assembled in part from Freesound [3], and the 600 source recordings of the environmental corpus were taken from Freesound. The model may therefore have encountered those very recordings during pretraining. This is not a hypothetical concern, and its size can be bounded by comparison with a corpus outside the pretraining data: on the environmental corpus CLAP’s advantage over the hand-specified descriptors is $+0.070$, and on DEAM, which is not sourced from Freesound, it is $+0.037$. Roughly half of the apparent advantage on the environmental corpus is attributable to overlap.

CLAP is also not a generally stronger representation. Within a corpus it reaches 0.787 against AST’s 0.794; on music-to-music transfer, the comparison that excludes the environmental corpus entirely, it reaches 0.620 against AST’s 0.619 and 0.614 for the hand-specified descriptors. Its advantage appears only in the comparisons that necessarily involve the corpus its pretraining data overlap.

Why this cannot be resolved with the available corpora. There is one environmental corpus in this work, so every cross-domain cell involves it. Genuine crossing and familiarity with the target therefore cannot be separated: the two readings — that CLAP crosses the boundary better, and that CLAP has seen the target — predict the same pattern in the data we have. We state plainly that we cannot distinguish them. The barrier claim in the main text accordingly rests on the three uncontaminated families, and CLAP is reported and flagged in Table 1 but used neither as support nor as refutation. Resolving it requires an environmental corpus that lies outside the pretraining data of the representation under test, which we were unable to obtain in a form suitable for these analyses; a second environmental corpus of any provenance would at least allow the comparison to be repeated without the overlapping one.

Relation to Extended Data Fig. 7. That figure is the earlier single-representation comparison, in which hand-specified descriptors were set against CLAP embeddings on two corpora. Its

headline — that the pretrained representation is stronger — is superseded by the four-family analysis in Fig. 1f and Table 1, which shows that the advantage is corpus-dependent and, on the environmental corpus, half attributable to overlap. It is retained because the contamination control it contains, the clean-corpus comparison on DEAM, is the source of the +0.037 quoted above.

S8 Withdrawn analyses

Each entry below was reported in an earlier version of this work and is no longer claimed. They are kept because a reader who encounters the earlier numbers should be able to find out what happened to them, and because several of the main-text analyses are constructed as they are in order to avoid repeating the error.

S8.1 Withdrawn quantities and inferences

Each entry states what was withdrawn, why, and what replaced it.

$\tau = 0.106$ as the between-person standard deviation of the slope for song-level electrodermal response. The profile-likelihood estimator treats each person’s slope variance as known, when in fact it is estimated from $n - 2$ degrees of freedom; the estimation noise has nowhere to go and is absorbed into τ^2 . Calibrated against a simulation with a known true value, the estimator returns 0.105 at 18 observations per person when the true τ is zero, and 0.039 at 44 observations. The reported 0.106, from a design with 18 observations per person, therefore sat *below* its own null. Across the 34 cells then in use, seven were in this position, all of them in that design. **Replaced by** a per-cell null built by parametric bootstrap from that cell’s own x vector, per-person observation count and residual scale with τ set to zero, and a variance-additive correction $\tau = \sqrt{\max(\hat{\tau}^2 - \hat{\tau}_{\text{null}}^2, 0)}$, verified against the known-truth simulation to a mean absolute error of 3.3% on $\hat{\tau}$. The per-cell nulls in that design run from 0.106 to 0.123, which a single global constant cannot cover. The corrected median for the electrodermal setting is 0.000, with 11 of 15 cells indistinguishable from their own null; the median n^* for that setting is taken over the five cells that retain a finite break-even (main-text Table 3). The full 33-cell table, with each cell’s null, correction and interval, is provided as Source Data.

“The confidence interval excludes zero, so individual differences are real and estimable”. That interval is for $\tau_{\text{eff}} = \sqrt{\tau^2 + \text{null}^2}$ and not for τ . Coverage was measured directly rather than argued: with a true τ of zero and 18 observations per person, only 16% of replicates cover zero, so 84% “exclude zero”. At that design the statement carries almost no information. **Replaced by** the corrected τ above. One companion reading from the same analysis survives and is retained — that all 15 interval upper bounds lie below break-even — because an upper bound on τ_{eff} also bounds τ , and is therefore conservative in the direction claimed.

The dose-response rank correlations at $n = 4$. Rank correlations of +0.632 and +1.000 for spectral smoothing, and −0.20 and −0.40 for transient injection, were computed on four

dose-averaged points. With $n = 4$ the smallest attainable two-sided p for a Spearman correlation is 0.083, so the statistic could not have reached significance whatever the data showed; averaging the 80 excerpts within dose also discarded the variation the claim is about. **Replaced by** a slope per excerpt across its four doses, over 80 excerpts, tested by Wilcoxon signed-rank against zero with a sign-flip permutation null, and reported together with the proportion of excerpts moving in a consistent direction. The replacement changes what can be claimed: spectral smoothing survives on both counts, whereas transient injection reaches significance as a mean shift while moving only 50% and 59% of excerpts consistently, and is reported as having no reliable per-excerpt effect (Boundary 2).

An EEG stimulus-level intraclass correlation of 0.892. Label leakage. The connectivity table was merged with a suffix rule that produced a column `stim_c` — the stimulus identifier itself — which was absent from the hard-coded exclusion list and was therefore treated as an EEG measure. A stimulus identifier has, by construction, perfect stimulus-level reliability. **Replaced by** 0.092, 95% CI [0.037, 0.142], obtained after a structural gate that does not depend on column names: any column with zero within-stimulus variance is a deterministic function of the stimulus and is dropped, which removed one column (157 measures $\rightarrow$ 156). The value is referred to a null built the same way as the observation — permute the stimulus labels, recompute all 156 intraclass correlations, retain the largest, 400 times — giving a selection-corrected $p = 0.020$. It is reported as a bound and not as an absence, and the comparison with the best-agreeing rating scale on the same trials (0.221, [0.155, 0.281]) is a 2.4-fold gap between non-overlapping intervals.

Required observations per person of 271, 650, $> 1,000$ and $> 1,000$. The simulation grid was too coarse — 60 replicates per cell and n extending only to 1,000 — to resolve the two weakest effects, which were reported as bounds. **Replaced by** 314, 564, 1,364 and 4,888 for $r = 0.40, 0.30, 0.20$ and 0.10 , from 500 replicates per cell with n extended to 5,000. The two strongest requirements moved by less than the Monte-Carlo error of the coarser run; the two weakest were resolution limits rather than estimates (main-text Table 2).

A requirement of 97–184 observations per person, presented as agreeing with an independent calculation. It was derived from the withdrawn $\tau = 0.106$, and the second calculation was not independent of the first: its target τ was the same constant, copied. **Replaced by** per-cell values read from the corrected table, giving $n^* = 67, 166$ and 318 in the three settings (main-text Table 3). The equal-budget surface, recomputed against the corrected per-cell values, separately gives 174 observations per person for the electrodermal setting with interval $[65, \infty)$. The infinite upper end is not a typographical artefact: the corrected lower bound on τ in that setting is zero, and no finite budget buys a quantity that may be zero.

A mechanistic account of the domain gap in terms of tonality. The prediction that tonal descriptors would degrade cross-domain transfer was recorded as passing on its direction alone. The magnitude accounts for about 4% of the gap, which does not support a mechanistic claim. **Replaced by** reporting the gap as a phenomenon with no mechanistic account. The

tonality result is reported instead as what it is: a targeted representation change that recovers a specific deficit, on one axis of one corpus.

This paragraph said something stronger in the preprint — that the tonality result was “an information-limited gap that a change of representation does close, and the pre-declared falsification test of the distinction that organises the paper”. Both halves are withdrawn. The distinction no longer organises the paper, so there is nothing for the result to falsify; the test’s necessary other half, that additional data does *not* close the happiness gap, was never measured; and no artifact predates the analysis for the criterion. See the Revision record.

Statements that multiple independent paths converge on a conclusion. Most rested on descriptor rankings shared between models, which Section S6 shows is not evidence of anything, and one rested on two paths that shared an input constant. **Replaced by** a single surviving instance: the three feature pipelines of the loudness control in Section S3, whose best correlations are 0.045, 0.049 and 0.072. The phrase is not used anywhere else in this work, and no other result should be read as claiming it.

Three further withdrawals are documented in full elsewhere in this document because they doubled as admission criteria: the conclusion that peripheral responses carry no reproducible stimulus-level component (Section S2), the cross-model attribution consensus (Section S6), and “best lag 6.0s” read as a latency estimate (Section S5).

S8.2 Analysis decisions that failed and were replaced

These were caught before publication rather than after. We list them because each was a decision that looked reasonable when made, returned a clean-looking result, and would have propagated a wrong conclusion had it not been caught.

Averaging attribution values across models. Values from a probability-output model and a log-odds-output model differ in scale by roughly an order of magnitude, so their mean is dominated by the second. Rules are now derived per model and required to agree.

A criterion window placed for the wrong component. The stimulus-locked control was first evaluated over 0–300 ms, chosen for a classical auditory N1. The response in these data peaks at 452 ms. Following the criterion would have led us to discard a valid dataset.

A null formed by moving in time. Temporally shifted nulls for the evoked response move the analysis window out of the quiet pre-stimulus period and into the response-collection period, which has a different noise environment. Equalising artefact handling between arms did not fix this; the sign-flip null, which does not move in time, does.

Leave-one-out cross-validation scored by correlation. Observed and null values both fall near -0.5 , because training-fold means are displaced from held-out values. The comparison is uninformative rather than merely biased; five-fold cross-validation with a per-participant permutation null replaced it.

A permutation that permuted nothing. In the stimulus-mapping recovery, the null initially permuted the rows of the profile matrix. The optimal assignment is invariant to row permutation, so the null distribution was identical to the observed value — its mean and the observation agreed to four decimal places, which is what exposed it. The corrected null permutes stimulus labels at the trial level and regenerates the profiles.

Feature columns defined by exclusion. Descriptor tables that also carry label columns were passed to models using an “everything except known metadata” rule. Newly added corpora carry additional label columns, which the rule silently admits as features. This recurred until it was fixed structurally; the leaked EEG intraclass correlation withdrawn above is the most consequential instance. Feature columns are now selected against the label file’s own column list, with a dimensionality assertion, and any column that is a deterministic function of the grouping variable is dropped regardless of its name.

A verdict printed before the number. One analysis script printed a fixed conclusion — that stimulus-level EEG reliability would not exceed about 0.10 — unconditionally, while the value it had just computed was 0.892. The contradiction raised no error because nothing compared the two. Verdicts are now derived from the computed value.

A significance threshold compared against the wrong null. Having corrected that leak, we then compared the resulting 0.092 against 0.040, the 95th percentile of the null for a *single* measure, and concluded it was significant. But 0.092 is the maximum over 156 measures. The correct null permutes stimulus labels, recomputes all 156 and retains the largest, which is the comparison reported in the main text.

S9 Extended descriptions of the main-text figures

The main paper’s captions are held to IEEE length. The descriptions below carry the methodological detail that would otherwise be lost: what each panel plots, which estimator produced it, which corpus pairs it rests on, and — where a panel’s reading changed in this revision — what the preprint said about it. Each description belongs to the main-text figure of the same number.

Figure 1 — the corpus boundary. Arousal models transfer between corpora of the same sound domain far better than across it, under every representation tested. **a** Generalisation stages for six regression algorithms (RF, XGBoost, GBDT, SVR, KNN, ElasticNet) over 122 spectro-temporal descriptors. Each line follows one algorithm from within-corpus cross-validation, through same-domain transfer (music $\leftrightarrow$ music), to cross-domain transfer (music $\leftrightarrow$ environmental); grey points are the individual corpus pairs behind each mean. Within-corpus values are GroupKFold-5 estimates grouped by source recording; transfer values fit on the whole of one corpus and are evaluated on the whole of another, with nothing refitted. Note that ‘grouped by source recording’ is a property of the group identifiers each corpus supplies, and on the environmental corpus those identifiers are one per clip: 613 of its 1,213 clips are mixtures of 30 source recordings, so GroupKFold there is arithmetically identical to KFold and near-duplicate mixtures span folds. Panels a-f use the corpus as distributed, for comparability with prior work; Table 1 of the main text gives the leak-free counterparts, on which the within-corpus reference falls from 0.744 to 0.646 and median cross-domain transfer rises from 0.043 to 0.129. **b** Cross-corpus transfer matrix, averaged over the six algorithms; the diagonal shows within-corpus cross-validation. Green indicates positive rank correlation, red negative. **c** Directional asymmetry. For each music corpus, transfer outward to the environmental corpus and inward from it, averaged over algorithms; the dashed line marks the same-domain mean (0.59). The inward direction is uniformly sealed ($\rho \leq 0.09$ for all three corpora), whereas the outward direction is not and increases from PMEmo (−0.04) through DEAM (+0.13) to Soundtracks (+0.24). Film soundtracks contain substantial atmospheric material -- sustained tones, texture, sound design -- which is a plausible account of that ordering but not one measured here. The barrier is not symmetric, and we report the asymmetry without an explanation for it. **d** Duration control. Music clips truncated from 30 s to a 6 s centred window, matching the environmental corpus; filled circles are 30 s, open circles 6 s. Same-domain transfer survives truncation (0.67 $\rightarrow$ 0.63) while cross-domain transfer stays at zero (0.03 $\rightarrow$ 0.05), excluding clip length as the cause of the collapse. **e** Leave-one-category generalisation inside the environmental corpus ($n = 1,213$), holding out each of the six Schafer soundscape categories in turn. The shaded band marks the mean cross-domain level. Generalisation to unseen content categories holds (mean $\rho = 0.52$; range 0.20 for quiet to 0.70 for human’, five of six categories above 0.33), so the collapse in **a–b** is specific to the domain border rather than to any content shift. **f** The same operation costed twice, under five representations. For each representation the corpus is swapped and nothing is refitted; the grey mark gives the performance lost when the new corpus is in the same domain,

the red mark the loss when it is not, both as a fraction of that representation’s own within-corpus performance, and the printed value is the gap between them in percentage points. Points are medians over the corpus pairs of **a**. For the three uncontaminated families the loss is 20–31% within the domain and 84–95% across it — hand-specified 20.2/93.9%, AST 22.0/84.2%, MERT 29.1/94.9%, Wav2Vec2 30.7/90.0%, a gap of 59 to 74 percentage points (Table 1). The result is where the collapse stands, not how high it is: every representation transfers between corpora and none crosses between domains, which disposes of the reading in which the embeddings simply fail to move between datasets. AST is the family that most ought to cross, since AudioSet contains large numbers of environmental categories; it raises cross-domain ρ from 0.047 to 0.125, a factor of 2.7 that is still a sixth of what the same representation reaches within a corpus. CLAP is drawn apart and greyed. Its pretraining set contains Freesound, the source of the environmental corpus, an overlap declared before any of these numbers existed. It is not a generally stronger representation — 0.620 on music-to-music against AST’s 0.619 and 0.614 for the hand-specified descriptors — and its lower barrier (57.3%) appears only in comparisons that necessarily involve the corpus its pretraining data overlap. With a single environmental corpus available, genuine crossing cannot be separated from familiarity with the target; the barrier claim rests on the three uncontaminated families, and CLAP is used neither as support nor as refutation. Corpora: DEAM (music, $n = 1,233$), PMEmo (music, $n = 717$), Soundtracks (music, $n = 360$) and Emo-Soundscapes (environmental, $n = 1,213$). Soundtracks (Eerola & Vuoskoski, 2011) differs from the other music corpora in collection site, decade, rating protocol (nine-point Likert scales over eight emotion terms rather than continuous valence–arousal sliders) and material selection (a balanced design of twelve target emotions $\times$ 30 excerpts). Its inclusion therefore tests whether within-music transfer reflects shared acoustic structure or merely a shared annotation tradition; transfer into it from the other music corpora is 0.56, and from the environmental corpus 0.09. Panels **a–e** rest on 12 ordered corpus pairs, each evaluated with the six algorithms: over the 36 same-domain values transfer averages 0.592 (s.d. 0.130) and over the 36 cross-domain values 0.054 (s.d. 0.161). Target is continuous perceived arousal rescaled to $-1 \dots 1$. All audio is loudness-normalised to -23 LUFS before feature extraction, which is an admission criterion for this figure rather than a cosmetic step (Fig. S1). Metric is Spearman ρ throughout. Source data are provided as a Source Data file.

Figure 2 — what adaptation recovers. Target-side labels recover about half of either corpus-swap gap, and label-free alignment recovers a quarter of the cross-domain one. **a** Fraction of the gap recovered by k labelled target-corpus examples, for both kinds of corpus swap. The gap runs from zero-shot transfer ($k = 0$, defined as 0%) to a model trained on the target corpus itself (100%), so the denominator is the target’s own within-corpus reference and not same-domain transfer — the question an engineer faces is how close labelling k examples gets to labelling all of them. Line and points, median over the six ordered corpus pairs of each kind; band, interquartile range across those pairs. Across domains: 19%, 31%, 44%, 57% and 71% at $k = 10, 25, 50, 100$ and 200. Same domain: 9%, 16%, 34%, 50% and 77%. The two curves differ in shape rather than in whether a price exists; the cross-domain curve rises faster at small budgets because its zero-shot starting point is much lower, and the same-domain curve overtakes it at 200. **b** Alignment without labels, with its sham control **split by validity**. Correlation

alignment (CORAL) recovers 25% of the cross-domain gap with no target labels. The sham aligns the source not to the target but to an unrelated third corpus, which by construction carries no target-specific information — except that with one environmental corpus available, the third corpus lies inside the *target’s* own domain on three of the six ordered pairs, so the sham is a valid negative control on only half of them. Separated, the valid sham recovers -11% and the invalid sham $+25\%$: the entire apparent sham effect is the invalid half. Subspace alignment is worse than doing nothing (-5%); the axis is not clipped at zero, because clipping would turn “actively harmful” into “small”. **c** CORAL decomposed. Matching per-dimension variance alone recovers 5%; adding correlation structure adds 20 points more, 80% of the total. That second step is target-specific by construction, which is the claim the pooled sham was supposed to test and could not. Every fraction is computed within an ordered corpus pair and only then pooled. Pooling first — averaging the correlations and taking one ratio against one median ceiling — mixes pairs whose baselines differ by more than the effect being measured. The environmental corpus here excludes its 613 mixture clips, which are combinations of 30 source recordings that the supplied group identifiers do not group; on the corpus as distributed they leak across folds and inflate both the target’s within-corpus reference and the apparent size of the gap. The estimator is ridge regression on the hand-specified descriptors. The 200-label point rests on five of the six cross-domain pairs and four of the six same-domain pairs: Soundtracks ($n = 360$) cannot give up 200 labelled examples and still be evaluated. Restricting every budget to the pairs present at all of them changes the cross-domain values by at most 2 points and the same-domain values by up to 5, and does not change the ordering. **What this figure corrects.** The preprint drew a single curve, filtered to the cross-domain pairs, and reported it in the abstract and discussion as the *same-domain* price while describing the cross-domain boundary as one on which no purchase could be found. This legend stated the correct condition; the prose did not. The preprint also reported that the return per doubling collapses after 100 labels, from a constant 14–16 points to 5. On the leak-free corpus the increments are 10, 14, 10 and 14 — no collapse, and the largest increment is the last. That claim is withdrawn, and with it any statement about where the curve saturates: it had not saturated at the largest budget tested. Metric is Spearman ρ ; all audio is loudness-normalised to -23 LUFS. Source data are provided as a Source Data file.

Figure 3 — the tonality result. A representation change recovers a specific deficit that more of the same descriptors does not reach. **a** Cross-validated Spearman ρ for eight affect axes of the Soundtracks corpus ($n = 360$), using 122 spectro-temporal descriptors alone (open circles) and with 39 tonal descriptors added (filled). Axes are ordered by gain; numbers give the gain. Blue marks the three axes for which musical mode is theoretically decisive (happiness, valence, sadness) — a grouping fixed a priori, not read off the result. The criterion stated for this test was that the gain concentrate on happiness, at more than twice the mean gain of the other seven axes and above 0.05 in absolute terms. It does, at $+0.132$ against a mean of $+0.045$, a ratio of 2.9. We can produce no artifact predating the analysis for that criterion and do not present it as preregistered. The ordering of gains follows mode-dependence: happiness, valence ($+0.083$) and sadness ($+0.077$) lead, while energy ($+0.012$) and anger ($+0.025$), which depend on level, roughness and rhythm, gain least. **b** Tonal descriptors alone against spectro-temporal descriptors alone. For happiness, 39 tonal descriptors ($\rho = 0.580$) outperform 122 spectro-temporal ones

(0.474); the deficit was one of representation rather than of dimensionality. **c** Mechanism check. Rank of five tonal descriptors among all 161, from random forest importance, for the happiness and tension axes (log scale). The two families dissociate: major-key strength and mode score rank 1st and 2nd for happiness but 3rd and 20th for tension, whereas interval consonance and sensory roughness rank 4th and 7th for tension but 117th and 76th for happiness. Mode carries perceived happiness; consonance and roughness carry perceived tension. The ordering was selected by the model from 161 descriptors, not imposed. The preprint presented this as a falsification test of a two-kind framework — an information-limited gap that a representation closes. That framing is withdrawn. The framework is withdrawn, so there is nothing to falsify; and the test’s necessary first half, that additional data does **not** close the happiness gap, was never measured, since no sample-size curve exists for any Soundtracks axis. What the result supports is narrower and still useful: a targeted representation change can recover a specific deficit, and which representation helps depends on which axis is being predicted. Bootstrap over out-of-fold predictions, resampled by stimulus, puts the happiness gain at +0.132 (95% CI [+0.059, +0.209]), its excess over the mean of the other seven axes at +0.087 ([+0.018, +0.159]) and the tonal-over-spectro- temporal margin on happiness at +0.106 ([+0.007, +0.208]); the last two lower bounds sit close to zero, and because the bootstrap resamples predictions without refitting the models the intervals are narrower than a full resampling would give. Six of six algorithms show a larger gain on happiness than the mean of the other axes. Tonality is not the mechanism of the domain barrier: adding tonal descriptors lowers music-to-environmental transfer by 0.021 (18 paired comparisons, $p = 0.038$ uncorrected), in the predicted direction but about 4% of a gap that runs from 0.591 to 0.115 on the leak-free environmental corpus. Interval-consonance weights are fixed a priori from the standard ordering of sensory consonance and are not fitted, so they add no degrees of freedom. All audio is loudness-normalised to -23 LUFS; cross-validation is GroupKFold-5 and each cell reports the best of six algorithms. Source data are provided as a Source Data file.

Figure 4 — audio editing. An acoustic edit moves model output, reliably per excerpt for only one of three interventions and not consistently across model and material. **a** Dose–response curves for three interventions applied to 80 source excerpts (40 per domain) at four dose levels, with dose 0 the unmodified control. Curves show the mean predicted probability of the low-arousal class under a model trained on the music corpus and a model trained on the environmental corpus; dotted lines mark the control level. The modelled quantity throughout is $P(\text{low arousal})$, so a rising curve means the edit made the excerpt read as **calmer**. Slowing spectral change raises the low-arousal probability in both domains; injecting transients lowers it in both. Envelope compression is retained as a failed manipulation. The curve is an average over 80 excerpts and cannot by itself distinguish an effect from a mean shift; **d** does that. **b** Manipulation check, run before any dose–response was read. Within-clip Spearman ρ between dose and the targeted descriptor; point size gives the fraction of clips moving in the same direction as the dose, and the shaded band marks $|\rho| < 0.5$. A target descriptor counts as moved only if $|\rho| \geq 0.5$ and at least 75% of clips move in the same direction, a criterion fixed in advance. Transient injection meets it on both of its target descriptors and spectral smoothing on two of three; envelope compression meets it on one of three and is carried through the figure as a failed manipulation rather than as a null effect. **c** Specificity diagnostic. The five descriptors moved

most by each intervention at maximum dose, in units of the training-set standard deviation. **d** Significance against reliability, per intervention and per domain model. Abscissa, the fraction of the 80 excerpts whose own slope across the four doses has the intended sign; ordinate, $-\log_{10}$ of a sign-flip permutation p on those 80 slopes. **The 50% line is the statistic's arithmetic floor, not a chance level:** sign consistency is computed against the sign of the observed median, so it cannot fall below 50% by construction, and under the null its median is 0.54. A point near 50% therefore describes an average over excerpts that disagree, not an effect the intervention has on audio in general, however small its p value, and values of 57–60% are close to the floor rather than above chance. Filled markers meet the reliability criterion used here — sign consistency $\geq 65\%$ and $p_{\text{perm}} < 0.01$ — and open markers do not; we can produce no artifact predating the analysis for that threshold and do not present it as preregistered. Spectral smoothing meets it in both domains (median slope $+0.015$ music, $+0.074$ environmental; sign consistency 70% and 71%; $p_{\text{perm}} = 1 \times 10^{-4}$ in both). Transient injection reaches significance ($p_{\text{perm}} = 1 \times 10^{-3}$ music, 8×10^{-4} environmental) at 50% and 59% sign consistency, at or barely above chance, and is therefore reported as a mean shift with no reliable per-excerpt effect and is not built upon. Envelope compression is uninterpretable here, having failed **b** ($p_{\text{perm}} = 0.50$ and 0.98). **These statistics pool, for each model, its own 40 training-domain excerpts with 40 from the other domain, and the preprint read the pooled result as showing that the direction of an edit crosses the domain boundary even though the predictions do not — "what fails is calibration rather than sign". Disaggregated, that does not hold.** The music model shows the effect on both kinds of material ($+0.099$ on music excerpts, $+0.012$ on environmental, both $p < 10^{-3}$). The environmental model shows it strongly on *music* excerpts ($+0.080$, 100% sign consistency) and not on environmental excerpts — the material it was fitted on, and the material the claim is about — where the median slope is -0.019 in the wrong direction at 57.5% sign consistency and $p = 0.073$. Three of four model $\times$ material cells move as intended and the fourth is the one the claim needs most, so the claim is withdrawn. Independently of the data it was incompatible with the metric: every transfer number in this work is a Spearman rank correlation, which is exactly invariant under any strictly monotone recalibration, so a purely calibrational deficit cannot move ρ at all. The perturbed models are binary random-forest classifiers of low- versus high-arousal, one per domain, and are not the regression models of Fig. 1. Panel **c** bounds what may be claimed: no intervention moves its target descriptor most. Transient injection moves `spec_contrast6_dmean` by 9.0 SD while moving `onset_rate` by 2.0 SD, ranking the target twelfth. Because the co-moving descriptors are themselves temporal-variability measures, these experiments support a claim about the variability *dimension* and not about any individual descriptor, and they are statements about model behaviour: no listener response to these edits was measured. An earlier version of this analysis averaged the 80 excerpts within dose and correlated the four resulting points; at $n = 4$ the smallest attainable two-sided p for a Spearman correlation is 0.083, so that statistic could not have reached significance whatever the data showed, and the excerpt-level test in **d** replaces it. All interventions are re-normalised to -23 LUFS afterwards so that loudness cannot drive the effect. Source data are provided as a Source Data file.

Figure 5 — response channel. Reference distributions, positive controls, and the comparison they license. **a** Null distribution of the number of descriptors shared by the top-20 SHAP sets of three models under 20 label permutations. The observed value (4) falls at the null mean (3.75), $p = 0.381$. Shuffled labels produce the same overlap as intact ones, so the statistic cannot separate signal from noise and no claim is built on it. **b** Cross-validated AUC under the same permutations, confirming that permutation destroyed the signal (null mean 0.496) while the intact labels reach 0.936. The failure in **a** is therefore a property of the statistic, not of the data. **c** Positive controls for two physiological pipelines addressing the same question. In BIRAFFE2 the habituation control is significant in the direction opposite to the established effect ($\rho = +0.113$ across trials, $p = 0.002$; skin conductance responses are expected to decline), and post-stimulus windows do not exceed random windows ($p = 0.267$); applied per participant, none of 94 passed both controls and habituation ran in the expected direction for only 35%, so no usable subset exists and that pipeline is withdrawn rather than reported. In ds002721 all three controls pass: relative occipital alpha exceeds frontal alpha in 31 of 31 participants ($p = 1.9 \times 10^{-9}$); blink-locked averages reach 164 μV frontally, 3.98-fold the occipital amplitude and far above random windows ($p = 9.3 \times 10^{-10}$); and a sound-onset response is present (**d**). Dashed line, $p = 0.05$. **d** Grand-average response at fronto-central electrodes (F3, Fz, F4, C3, Cz, C4), time-locked to sound onset, $n = 31$. Dark band, s.e.m.; grey band, the 95th percentile of a subject-wise sign-flip null with max-statistic correction across the epoch; green shading, the interval exceeding it (372–588 ms). Peak $|t| = 8.78$ at 452 ms, $p = 0.0001$ (10,000 permutations). The pre-stimulus interval is not significant ($p = 0.54$), excluding a slow drift. No classical N1 is present; the stimuli have sharp onsets (median 25 ms to half-amplitude, Supplementary Information), so the absent fast component is attributed to playback-latency jitter, which limits millisecond-scale analysis but not trial-wise spectral measures. **e** Stimulus-level intraclass correlation for eight self-report scales and 156 EEG measures, computed on the same 31 listeners and with the same estimator after within-listener z-scoring. The two sets are **not** computed on identical trials, as the preprint stated: the EEG measures use 1,240 trials over 307 stimuli and the self-report scales 1,080–1,121 trials over 285–298 stimuli. Points are individual measures, vertical bars the median. The EEG set spans four families: regional band power (26), magnitude-squared coherence between region pairs (50), imaginary coherency (50, robust to volume conduction) and electrode-pair asymmetry (30); coherence and asymmetry are included because the original analysis of these recordings located its effect there, and restricting the set to band power would test the conclusion on a representation that excludes the effect in question. Coherence is the strongest EEG family (maximum intraclass correlation 0.092, 95% CI [0.037, 0.142], split-half 0.51) and band power the weakest (0.040, split-half 0.17), against 0.221 [0.155, 0.281] for the best-agreeing self-report scale on the same trials — a 2.4-fold gap between two intervals that do not overlap. The figure states one requirement in two settings: a statistic is uninterpretable without the distribution it would take under the null (**a**, **b**), and a negative result is uninterpretable until the instrument is shown to detect a known effect (**c**, **d**). Panel **e** is the comparison those controls license, and it is reported as a bound rather than as an absence. The EEG value is the maximum of 156 measures and is biased upward for that reason, so it carries a null built the same way: permuting stimulus labels, recomputing all 156 intraclass correlations and taking the largest, 400 times, places the observed

value at $p = 0.02$, while Benjamini–Hochberg across the same 156 does not pass it (smallest $q = 0.16$). The self-report value is also the maximum of a selection, but over 8 candidates rather than 156, so the two point estimates carry unequal selection bias in the direction that favours the EEG side; the comparison is conservative with respect to the conclusion drawn from it, but the asymmetry should be stated. The two corrections disagree because the value sits between them — at rank one the Benjamini–Hochberg threshold is the Bonferroni threshold — and 0.092 is close to the 0.10 this design resolves with 80% power, so the interval is consistent with values from near zero up to about a tenth of the response variance. What the data support is the comparison and not a zero: whatever stimulus-specific structure the EEG carries is less than half of what the listeners’ own ratings carry on the same trials. This is a statement about the component shared across listeners, not about whether the brain responds to sound — the onset response in **d** is large and highly reliable; what is weak is the differentiating component. An earlier version of panels **c–e** was built on an electrodermal analysis that was retracted when its pipeline failed the controls now shown in **c**, and an earlier stimulus-level value of 0.892 was a leak — the stimulus identifier itself had entered the measure set — and is withdrawn. Both are documented in the Supplementary Information. Source data are provided as a Source Data file.

Figure 6 — the individual-level design requirement. The individual-level question is a design problem rather than a null result, and its price depends on the setting.

a Detection measured in real physiology. In an independent corpus of 75 listeners with about 240 auditory events each, the evoked response to a tone was tested per person by a sign-flip permutation test on the difference between real cues and randomly placed anchors. Each point gives the proportion of listeners in whom the response was detected when only n of that listener’s observations were used (40 resamples per point). Three channels are shown, each with the analytic power curve for its measured effect size overlaid as a pale band; empirical and analytic differ by 0.015 on average. The dotted line is the false-positive rate obtained by sign-randomising the same differences, which preserves the noise and removes the effect; it stays at nominal (0.03–0.07) across the whole range, so the rise in the curves is detection and not false positives. The shaded band marks what a laboratory session supplies. Within it, an effect plainly visible in the group average ($-5.6 \mu\text{V}$ at the N1, group $p = 0.0002$ in all three channels) is recovered in 53% of individuals from EEG, 39% from pupil diameter and 23% from heart rate. **b** Probability of detecting a within-person association as a function of the observations available per person, for four true association strengths. Synthetic data were generated from the empirical feature covariance of the EEG corpus (Ledoit–Wolf shrinkage) with an association of known strength injected, then passed through the identical pipeline, criterion and per-person permutation test used for the real data (500 simulations per cell, 100 permutations each, n from 30 to 5,000). Within the laboratory-session band, power is 0.10 for $r = 0.40$, 0.072 for $r = 0.30$, 0.06 for $r = 0.20$ and 0.04 for $r = 0.10$. The $r = 0.10$ curve is not monotone below $n = 250$; the Monte-Carlo s.e. is 0.009 to 0.015 per cell and the curve is included for the floor it establishes, not for its shape. **c** Observations per person required for 80% detection, measured and simulated on one axis, against three supply levels. Every requirement is resolved by this grid; the two weakest effects, which a coarser run could only bound at $> 1,000$, now stand at 1,364 ($r = 0.20$) and 4,888 ($r = 0.10$). Dotted lines give the supply from this corpus (30), comparable public corpora

(60), and a single overnight recording sampled once per minute over eight hours (480). Heart rate, the signal a consumer device records, requires roughly four times as many observations as scalp EEG (233 against 57). **d** When per-person calibration begins to pay, in three settings. For each descriptor–outcome cell the between-person standard deviation of the acoustic slope was estimated by restricted maximum likelihood (filled circle, median over that setting’s cells) and compared with a null built by parametric bootstrap from that cell’s own design (cross, median over the same cells). The null is plotted rather than subtracted, because a between-person spread estimated from few observations per person is biased upward: at 18 observations per person the estimator returns 0.11 when the true value is zero, and a reader who cannot see that cannot judge the physiological row. Break-even, n^* , is the number of observations per person at which the corrected spread exceeds the error in estimating it. Urban soundscape ratings (42 observations per person, 8 cells): estimate 0.134 against a null of 0.013, $n^* = 67$, and 2 of 8 cells sit above break-even at the observation count the corpus supplies, both on the eventfulness axis. Two qualifications attach to those two cells and neither appeared in the preprint. The corpus’s own manipulation check passes for the pleasantness axis ($p = 4 \times 10^{-6}$) and **not** for eventfulness ($p = 0.38$), so the axis carrying both crossings has no passing positive control; and no out-of-sample personalisation test exists for this corpus, so “above break-even” here is a property of a variance model rather than a demonstration that a per-person slope beats the population slope. Music ratings (44 per person, 10 cells): 0.074 against 0.000, $n^* = 166$, none across. Music electrodermal responses (18 per person, 15 cells): 0.106 against a null of 0.114 — the estimate does not exceed its own noise — with $n^* = 318$ taken over the 5 cells in which it is finite, 11 of the 15 indistinguishable from their own null, and none across. Both kinds of estimate in **a–c** are lower bounds, for different reasons. The simulation assumes multivariate normality, a linear association and within-person stationarity, each of which favours detection. The measurement uses a deliberately plain pipeline — a fixed-window mean over one electrode cluster, no independent component analysis and no spatial filtering — so a dedicated pipeline would do better; it is, however, representative of what a non-specialist pipeline achieves. The extrapolation behind **c** was validated out of sample: predicting detection at n from an effect size estimated on a disjoint half of the same person’s trials matches the observed rate to within 0.03 on average. Panel **d** is a different quantity from **a–c** — break-even for per-person calibration, not detection of a within-person association. The three settings in **d** differ in corpus, descriptor set and outcome axis simultaneously (ISO-532 psychoacoustic descriptors against ISO soundscape axes; spectro-temporal and tonal descriptors against valence and arousal; the same descriptors against electrodermal measures), so the spread between them cannot be attributed to any one of the three and they are not compared against one another. Where the direct out-of-sample test was run it points the other way from the model: fitting each listener their own slope loses to the population slope in 15 of 15 electrodermal cells (-0.109 to -0.072 at 18 observations per person) and in every music-rating cell at 18, 30 and 50 observations per person, turning positive in 4 of 10 cells only at $n = 100$. Two earlier claims from that analysis are withdrawn and do not appear here: a between-person spread of 0.106 for the physiological setting, which sat below its own null; and the reading of a confidence interval excluding zero as evidence of individual differences, since that interval is for the composite of true spread and null and, at 18 observations per person with

a true spread of zero, excludes zero on 84% of occasions. Read together, **a–c** place the shortfall at roughly a factor of twenty and locate it in duration and repetition rather than in sample size: the quantity in short supply is observations within a person, which recruiting more participants does not provide. Source data are provided as a Source Data file.

S10 Supplementary figures

The eight figures below support specific steps in the main text. Figs. S1–S2 concern the admission criteria and framing decisions that precede Boundary 1; Figs. S3–S5 concern the models and the selection routes used to establish that transfer must be measured; Fig. S6 is the individual-level analysis at Boundary 3; Fig. S7 is the earlier single-representation comparison superseded by Table 1 of the main text (Section S7); and Fig. S8 is the descriptor-level sign analysis behind the statement that only level-independent descriptors keep their direction across the domain boundary.

Fig. S9 is the four-route feature-selection comparison, which appeared as a main figure in the preprint and is moved here because it is a methodological precondition rather than a result: the route that tests within-corpus significance returns, at a rate below chance, precisely the descriptors that do not survive the domain boundary, which is why every boundary in this work is crossed and measured rather than predicted from within-corpus attributions.

A further main figure of the preprint, a four-boundary summary, is withdrawn rather than renumbered. It encoded the budget/information dichotomy graphically and cannot be corrected, only removed; the reasons are set out below.

Withdrawn. The preprint’s summary figure placed the four boundaries on two axes — the fraction of an attainable ceiling that survives each crossing, and the price of buying the gap back — with colour carrying a diagnosis: budget-limited where target-side observations close the gap, information-limited where they do not. It is withdrawn rather than corrected, for three independent reasons. The diagnosis it encoded is withdrawn: the boundary the figure coloured information-limited is the one whose budget curve the paper reports, and that curve recovers 57% of the gap at 100 target labels. Panel **a** placed four incommensurable ratios on one axis labelled as a fraction of an attainable ceiling. The corpus row is transfer against a within-corpus model reference; the response-channel row is one intraclass correlation against another, which is a ratio of reliabilities and not of achieved performance; the individual row is observations supplied against observations required; and the synthesis row was drawn at 1 and 0 by construction and was never a measured fraction at all. Panel **a**’s electrodermal row additionally used the wrong corpus’s observation count as its numerator. Nothing in the paper now needs a figure of this kind. The quantities it summarised are reported per boundary, in the units each was measured in, and the ledger of what improved, what did not improve under the routes tested, and what the evidence cannot decide is given as text in the Discussion.

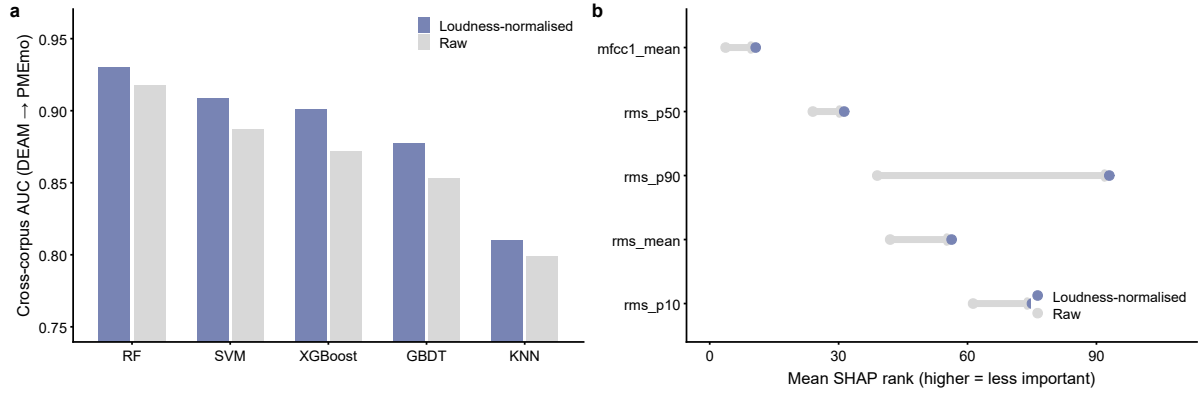


Figure S1: Loudness acts as a corpus fingerprint rather than as signal. **a** Cross-corpus AUC (DEAM → PMEmo) for five classifiers before and after loudness normalisation to -23 LUFS. Normalisation raises transfer for every model, from a mean of 0.866 to 0.886. **b** Mean SHAP rank of the five level-dependent descriptors before and after normalisation; arrows run from raw to normalised. All five lose importance, with **rms_p90** falling from rank 39 to rank 93. Removing absolute level improves rather than degrades cross-corpus transfer, which is the signature of a corpus-specific confound: mastering practice differs systematically between corpora. Because every transfer number in this work is computed on normalised audio, this control is the admission criterion for the domain barrier, and the obvious objection — that normalisation creates the barrier by discarding a genuinely informative dimension — was tested separately. Three feature pipelines that share no data, no intermediate table, no constant and no estimate — normalised descriptors, descriptors computed with loudness retained, and the 6,373-dimensional ComParE set extracted from the original recordings by the corpus authors — put the best cross-domain transfer at $p = 0.045$, 0.049 and 0.072. The spread of 0.027 is below the 0.05 threshold fixed before the test, and loudness alone contributes only +0.059. This is the only result in this work for which agreement between separately derived pipelines is claimed; no other comparison here meets that condition. Source data are provided as a Source Data file.

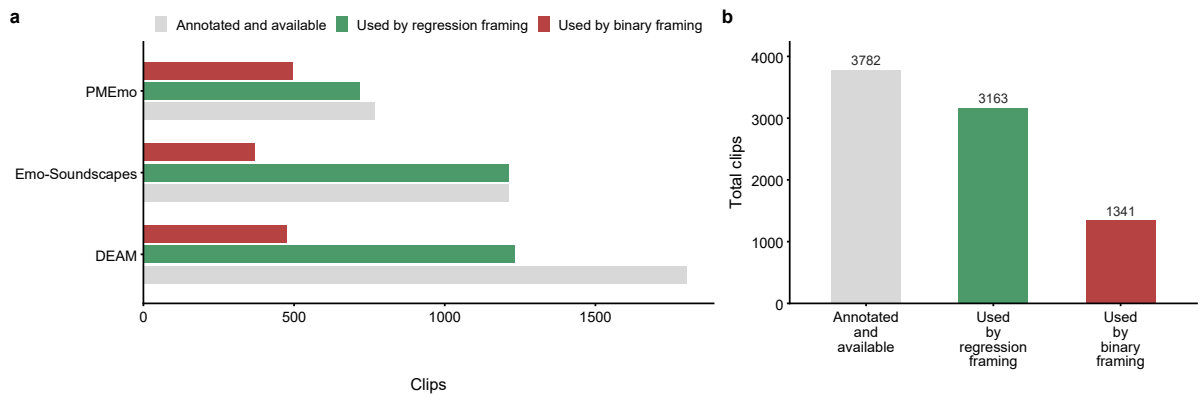


Figure S2: Regression framing recovers the discarded middle of the arousal range. **a** Clips available, used under the regression framing, and used under the original binary framing, per corpus. **b** Totals across corpora. The binary framing selects the two extremes of the arousal axis and discards the middle, retaining 1,341 of 3,782 annotated clips. The regression framing uses 3,163. The discarded clips carry the threshold information that a rule library requires. Source data are provided as a Source Data file.

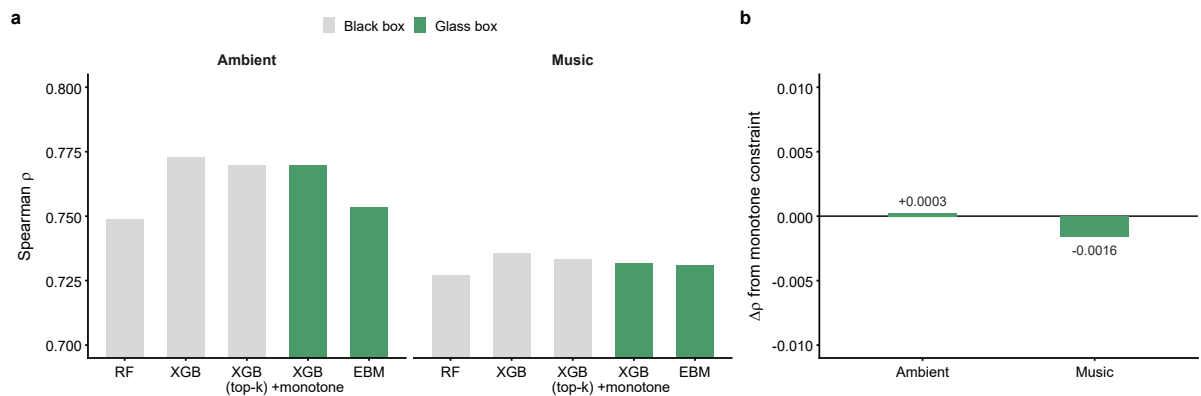


Figure S3: Glass-box models cost almost no accuracy. **a** Spearman ρ under GroupKFold-5 for black-box and glass-box models in both domains. **b** Change in ρ from imposing monotone constraints on the seven cross-domain-invariant descriptors, relative to the same feature subset without constraints. Monotone constraints cost +0.0003 in the ambient domain and -0.0016 in the music domain — replicated at zero cost in both. The monotonicity assumed by threshold rules is therefore supported by the data rather than imposed on it. The fully additive EBM trails the best black box by 0.019. Source data are provided as a Source Data file.

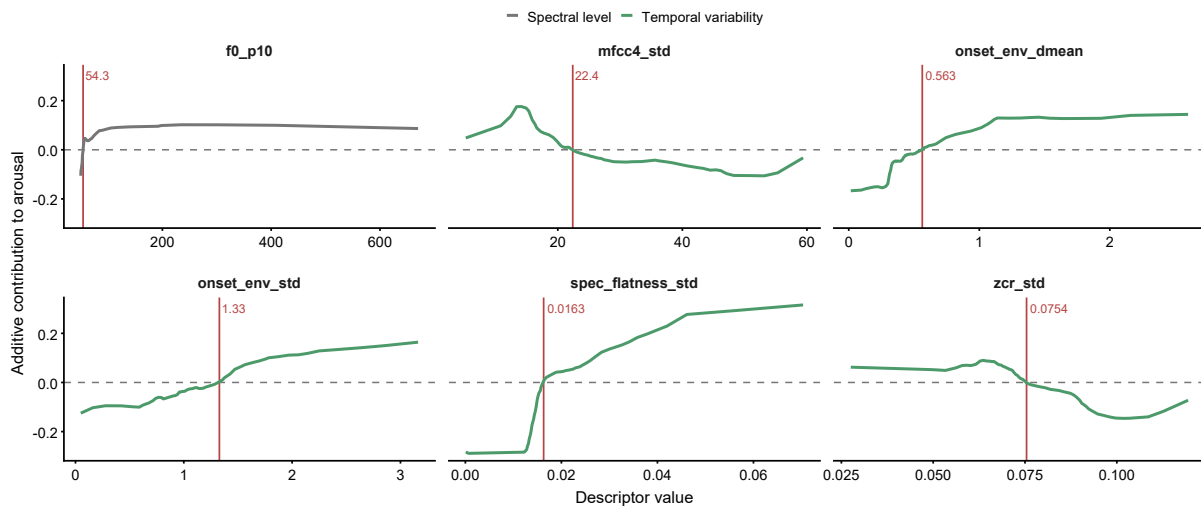


Figure S4: Explainable Boosting Machine shape functions give thresholds without post-hoc attribution. Additive contribution to predicted arousal as a function of descriptor value for the six most important terms in the ambient-domain EBM ($n = 1,213$, interactions disabled). Red lines mark the zero crossing, printed above each panel; these are the thresholds a rule library reads directly, with no attribution step. Five of the six terms are temporal-variability measures. Source data are provided as a Source Data file.

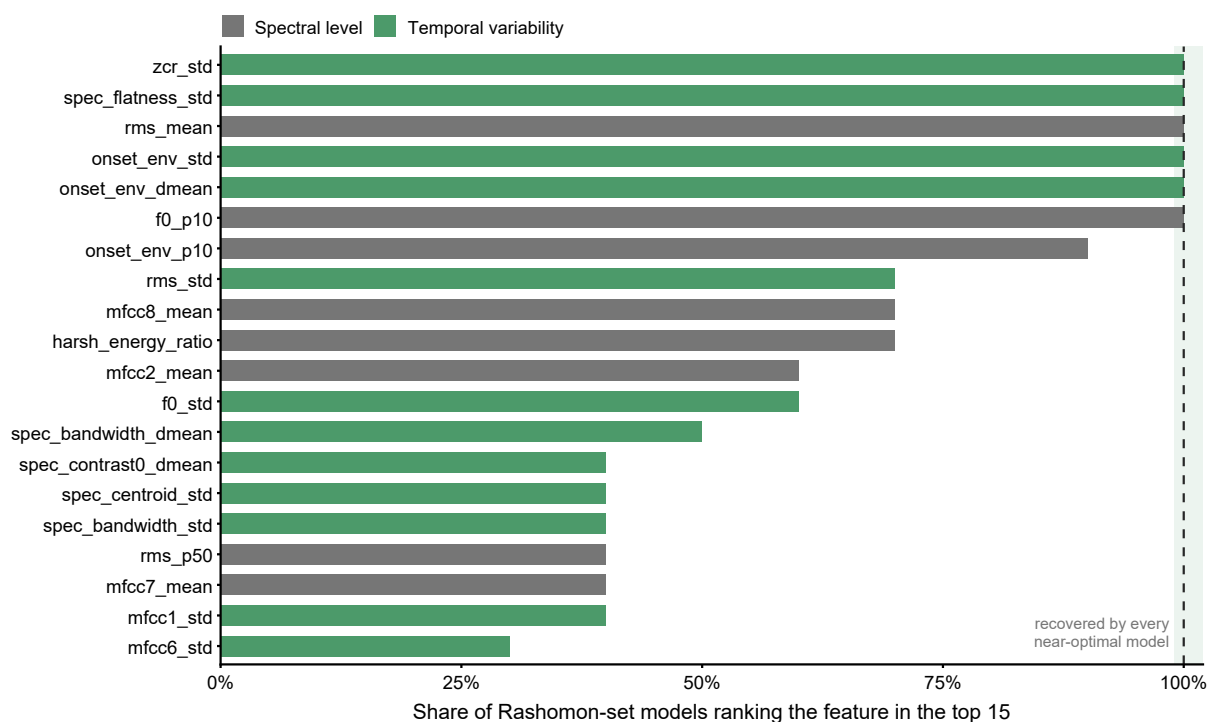


Figure S5: Descriptor importance is stable across the Rashomon set. Share of near-optimal models ranking each descriptor in their top 15. The Rashomon set comprises the 10 of 60 sampled models within 0.02 Spearman ρ of the best ($\rho \in [0.823, 0.843]$); models were sampled over algorithm family, depth, regularisation and subsampling, and importance was measured by permutation importance on a held-out fold. Six descriptors are recovered by every near-optimal model, four of them temporal-variability measures. Reporting importance from a single fitted model would have left this multiplicity untested. Source data are provided as a Source Data file.

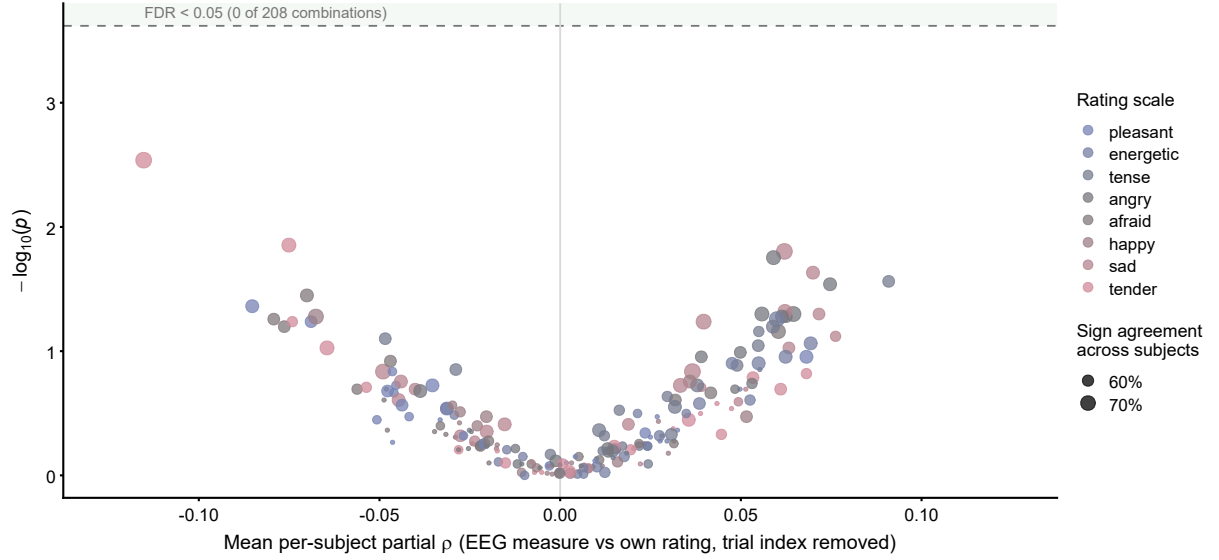


Figure S6: No EEG measure tracks self-reported emotion across listeners at the effect size this design can resolve. Each point is one of 208 combinations of an EEG measure (26) and a rating scale (8). The abscissa gives the mean across the 31 listeners of the within-listener partial Spearman ρ between that measure and that listener’s own trial-by-trial rating, with trial index partialled out; both EEG and ratings can drift over a session, and a shared drift would otherwise manufacture a correlation. The ordinate gives the evidence against a zero group mean (Wilcoxon signed-rank). Point size is the fraction of listeners whose sign matches the group mean; colour is the rating scale. The dashed line marks the Benjamini–Hochberg threshold at $\text{FDR} < 0.05$. No combination crosses it (largest $|\text{mean } \rho| = 0.115$; median 0.028; the group mean’s sign is shared by a median of 55% of listeners), and the distribution is symmetric about zero as expected under the null. This is a bound and not an absence. Listeners contributed a median of 30 rated trials each, and the design simulation (Fig. 6b) shows that at 30 observations per person a within-person association of $r = 0.40$ would be detected on 10% of occasions and one of $r = 0.20$ on 6% — that is, at chance. The panel therefore establishes only that no large within-person association is obvious, and supports no conclusion about associations below approximately $r = 0.5$. The group-level result in Fig. 5e does not share this limitation, because it pools 31 listeners at the level of the stimulus. This figure is computed on the pipeline validated in Fig. 5c,d, which is what makes the negative interpretable at all; independent-component removal of ocular artefacts, used here as a sensitivity manipulation check, increased detection of the known effect while further reducing the effect of interest. An earlier version used an electrodermal analysis that was retracted when its pipeline failed the same controls. Source data are provided as a Source Data file.

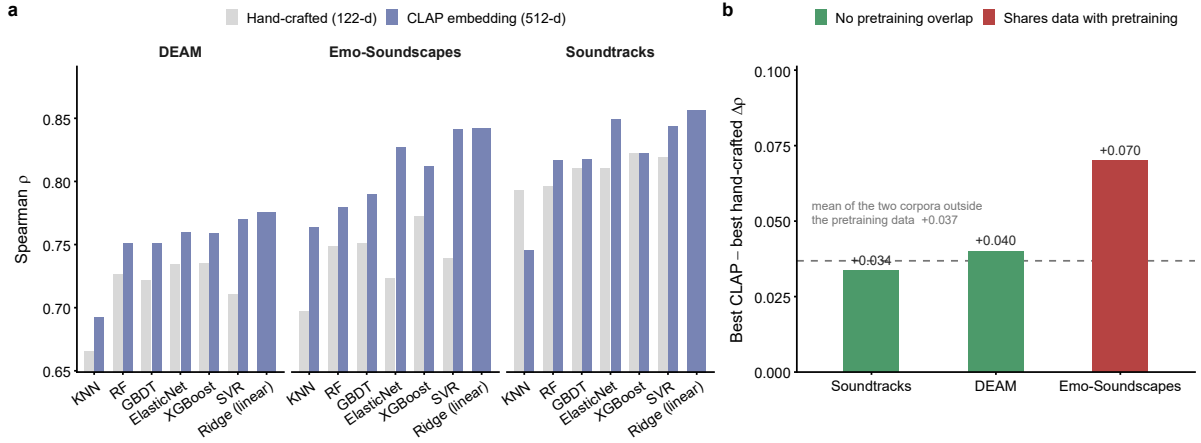


Figure S7: A pretrained representation gains most on the one corpus that overlaps its pretraining data. Best cross-validated Spearman ρ for 122 hand-crafted spectro-temporal descriptors and for 512-dimensional pretrained audio embeddings, on three corpora (best of six algorithms, GroupKFold-5, loudness-normalised audio). Embeddings win on all three: +0.040 on DEAM (0.736 $\rightarrow$ 0.776), +0.034 on Soundtracks (0.823 $\rightarrow$ 0.856) and +0.070 on Emo-Soundscapes (0.773 $\rightarrow$ 0.843). The advantage does not require a strong downstream model — on embeddings, cross-validated linear ridge regression is competitive with the tuned tree ensembles — so it is carried by the representation and not by the estimator. The two music corpora have no overlap with the embedding model’s pretraining data and give the two smaller margins; the corpus that derives from the public sound repository appearing in that pretraining data gives the largest. We report the music-corpus figure, +0.037, as the estimate that generalises and leave the difference unattributed. It cannot be assigned to pretraining overlap here: the one corpus sharing data with pretraining is also the only environmental corpus in the comparison, and both clean corpora are music, so overlap and sound domain are collinear. The larger margin is equally consistent with the embedding model simply being stronger on environmental sound. Separating the two accounts would require an environmental corpus outside the pretraining data, or a music corpus inside it. This figure concerns within-corpus accuracy only. Whether a pretrained representation crosses the domain boundary is a separate question, answered for four pretraining paradigms in Fig. 1f and Table 1, where none of them does. Source data are provided as a Source Data file.

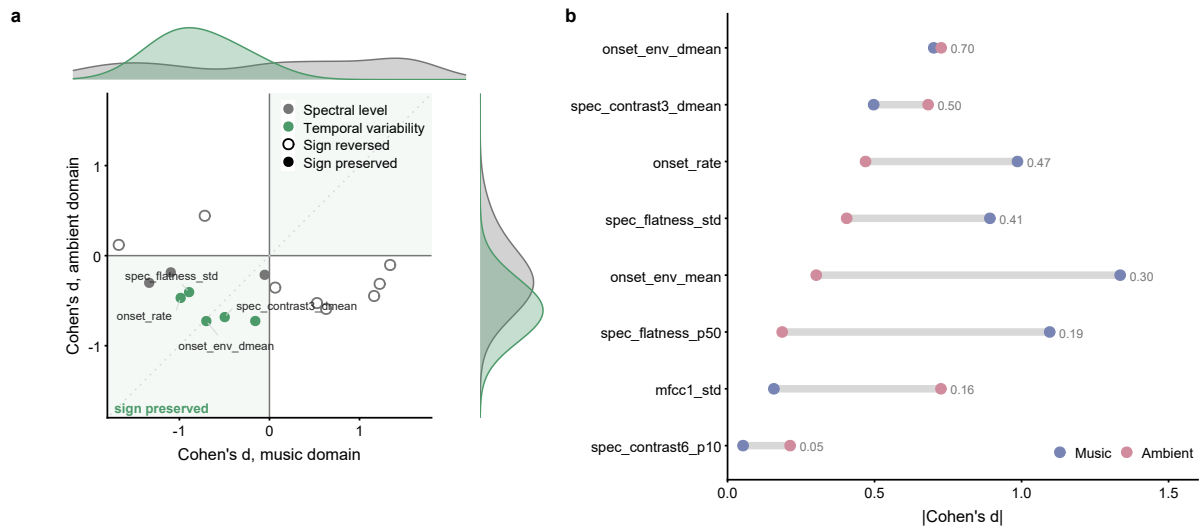


Figure S8: Only temporal-variability descriptors keep their direction across sound domains. a Standardised effect size (Cohen's d) of each FDR-significant descriptor in the music domain versus the environmental domain, with marginal densities by feature class. Shaded quadrants mark sign preservation; filled circles keep their sign across domains, open circles reverse it. The dotted diagonal is the identity line. Descriptors with $\min|d| > 0.35$ are labelled. **b** The eight sign-preserving descriptors, ranked by the weaker of their two effect sizes (printed at the right of each row). Circles give $|d|$ in each domain and the connecting bar gives the gap between them. All eight are negative in both domains: lower descriptor value corresponds to lower arousal, whether the sound is music or an environmental recording. The descriptors that reverse are without exception measures of absolute spectral level — the property on which the two domains differ most. Feature classes are assigned by statistic type — descriptors ending in `_std` or `_dmean`, plus `onset_rate`, are temporal-variability measures; all others are spectral-level measures. Significance was established per domain by per-feature permutation testing (200 permutations) with Benjamini–Hochberg control at $q = 0.10$. This analysis is the second of the four selection routes in Fig. S9, and is the only one that tests the boundary directly. Source data are provided as a Source Data file. This analysis appeared as a main figure in an earlier version; it was moved to the Supplementary Information when the price of the first boundary (Fig. 2) took the main-figure slot.

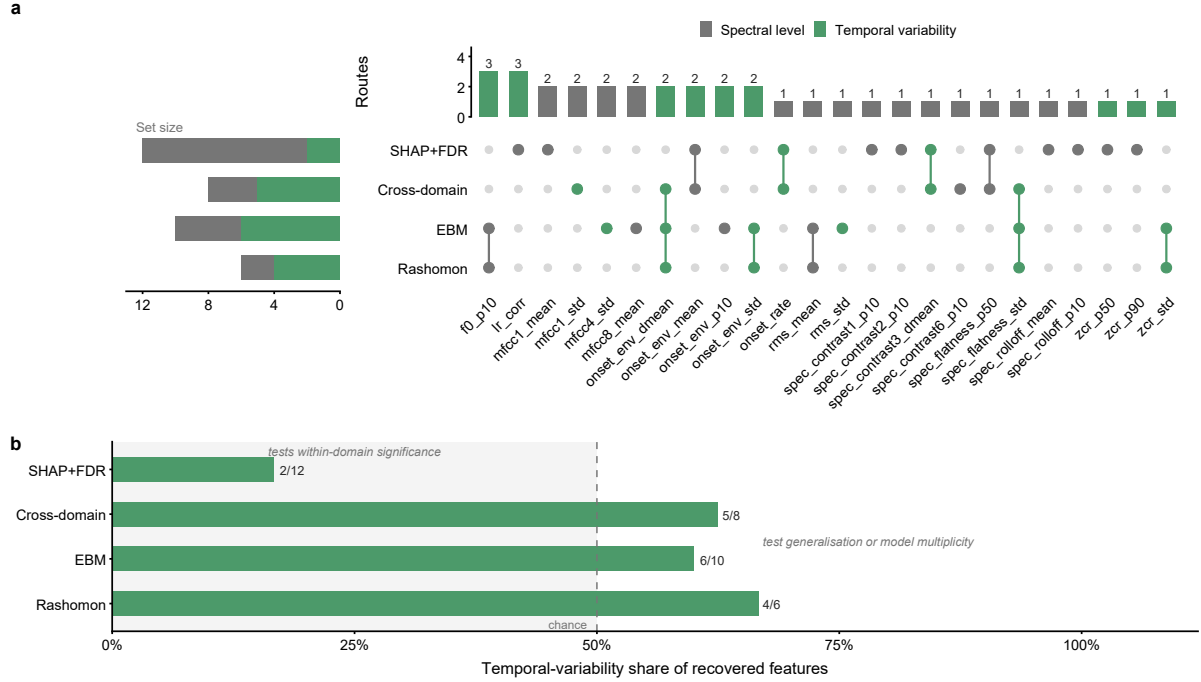


Figure S9: Feature-selection routes recover different descriptor classes depending on what they test. **a** UpSet-style recovery pattern. The matrix marks which of four selection routes recovered each descriptor; connected points indicate a descriptor recovered by more than one route. The top bars count routes per descriptor and the left bars give each route's set size, split by feature class. The four routes are: SHAP importance with per-feature permutation FDR ($q = 0.10$, 200 permutations); preservation of the sign of the cross-domain Cohen's d ; native shape-function importance from an Explainable Boosting Machine; and stability across a Rashomon set of 10 near-optimal models sampled from 60 ($\rho \geq \rho_{\max} - 0.02$). **b** Share of each route's recovered set that consists of temporal-variability descriptors, against the 50% chance line. The route that tests within-domain statistical significance returns 2 of 12 (17%), below chance; the three routes that probe generalisation or model multiplicity return 5 of 8 (63%), 6 of 10 (60%) and 4 of 6 (67%). The four routes differ in statistical principle — post-hoc attribution, effect size, additive shape functions, and model multiplicity — not in hyper-parameters, so **b** is a dissociation and not a shortfall of tuning. The route that is standard in the field returns, at a rate below chance, precisely the spectral-level descriptors that Fig. 1 and Fig. S8 show do not survive the domain boundary. What the panel licenses is therefore a negative: no attribution method here supports a statement about what will transfer, which is why every boundary in this work is crossed and measured rather than predicted from the descriptors a model relies on inside its own corpus. The companion requirement — that an apparent overlap between attribution sets be read against the overlap that shuffled labels produce — is in Fig. 5a,b. Source data are provided as a Source Data file.

S11 Methods detail held out of the main text

IEEE caps a regular paper at 18 formatted pages and places no limit on supplementary material. The material below is the part of the Methods that a reader needs in order to reproduce the analyses but not in order to judge the result, and it is here for that reason alone.

S11.1 The 39 tonal descriptors

Harmonic-percussive separation ratio; a key-invariant chroma profile obtained by estimating the tonic through correlation of the mean constant-Q chroma vector with Krumhansl–Kessler major and minor profiles [4] and rotating the vector so that the tonic occupies index 0; major-key strength, minor-key strength, their difference (mode score) and key clarity; chroma flux and chroma entropy; six tonal-centroid dimensions with standard deviations; an interval-consonance measure formed by weighting each pitch-class pair by the consonance of its interval; sensory roughness computed over the strongest spectral peaks per frame after Plomp and Levelt [5]; and fundamental-frequency median, interquartile range and voiced fraction.

The interval-consonance weights are fixed a priori from the standard ordering of sensory consonance (unison > perfect fifth > perfect fourth > major third and sixth > minor third and sixth > major second and minor seventh > minor second and major seventh > tritone) and are not fitted, so they contribute no degrees of freedom.

S11.2 Response measures in ds003690

Three measures were fixed before analysis, one per channel.

EEG. Mean fronto-central amplitude over 80–130 ms after cue onset, across nine electrodes, mastoid reference, 0.1–30 Hz, with epochs exceeding $\pm 150 \mu\text{V}$ rejected. A secondary EEG measure — the difference between 170–230 ms and 80–130 ms — was also specified. Inspection of the grand-average waveform showed the later window to fall on the contingent negative variation rather than on a P2, as is expected in a cued task, so the single-window measure was adopted as primary and the secondary is reported alongside it. Both are fixed-window means, for which the sign-flip null is exact; peak-to-peak measures are not, being positive under the null by construction.

Cardiac. Mean change in instantaneous heart rate over 1–5 s relative to a 3 s pre-cue baseline, with R peaks detected on a 5–35 Hz band.

Pupil. Mean change in pupil diameter over 0.5–2.5 s relative to a 1 s baseline, with blink samples interpolated and epochs exceeding 30% interpolation rejected.

Random anchors rather than a fixed temporal shift were used because the inter-cue interval leaves no quiet window of the required length; the difference also removes a monotone within-run decline in heart rate that the anchor arm revealed. The trial-count curve subsamples n of a participant’s pairs, repeats the identical test 40 times per value of n , and reports the mean detection rate; the accompanying false-positive rate applies the same procedure to sign-randomised differences, which preserves the noise and removes the effect. Extrapolation was validated out of sample by estimating each participant’s effect size on a random half of their trials and predicting detection on the other half. Heart rate and pupil diameter were analysed separately by task, a

stratification fixed on design grounds: the active-task response window contains the imperative tone and the button press, the passive block contains neither, and the two differ eightfold in trials per person.

S11.3 Design simulation

Feature covariance was estimated from the artefact-corrected EEG measures by Ledoit–Wolf shrinkage [6]. For each combination of true association strength ($r \in \{0.10, 0.20, 0.30, 0.40\}$) and observations per person ($n \in \{30, 60, 120, 250, 500, 1000, 2000, 5000\}$), 500 datasets were generated from that covariance with a sparse linear association injected and scaled to the target strength, and each was analysed with the identical pipeline, criterion and permutation test used on the real data (100 permutations per dataset). Reported power is the proportion of datasets in which the association was detected at $p < 0.05$. An earlier run at 60 datasets per cell and $n \leq 1,000$ could only bound the two weakest effects at $> 1,000$; extending the grid resolves them at 1,364 and 4,888, and the two strongest requirements shifted by less than the Monte-Carlo error of the coarser run (271 to 314, and 650 to 564).

The simulation is optimistic by construction. Multivariate-normal generation understates the heavy tails of real measurements; the injected association is linear; and within-person stationarity is assumed, which is least tenable over the long recordings the result is used to discuss. Required sample sizes are lower bounds.

S11.4 Per-listener decoding in ds002721

Five-fold cross-validation with standardisation and principal-component reduction to eight components fitted within each training fold, followed by ridge regression with a cross-validated penalty. The evaluation metric is the rank correlation between out-of-fold prediction and observation. This metric is biased downwards under cross-validation at small n — training-fold means are displaced from held-out values — so it is never compared against zero but always against a null formed by permuting that listener’s own labels through the identical pipeline (200 permutations). Leave-one-out cross-validation was tried first and abandoned: it produced observed and null values both near -0.5 , rendering the comparison uninformative.

S11.5 Artefact handling

For the main analyses no epoch rejection was applied, so that the stimulus and null arms receive identical treatment. In the sensitivity analysis, independent-component analysis (15 components) was fitted per run and components correlated with the frontopolar channels, used as ocular proxies, were removed (median one component per run). All other parameters were held fixed, so that any change is attributable to artefact removal alone.

S11.6 A mis-specified control criterion, and what it would have cost

Control 4 of the four dataset-level EEG controls initially failed under a criterion comparing evoked global field power in a 0–300 ms window against temporally shifted nulls. Two problems were identified. The window was placed for a classical N1 while the response in these data peaks

at 452 ms; and any temporally shifted null moves the analysis window into the response-collection period, which has a different noise environment, because stimulus onsets fall in the quietest part of the trial structure. The sign-flip null avoids both by not moving in time. We report this because the mis-specified criterion would, if followed, have led us to discard a valid dataset.

S12 Corpus provenance, ethics and licence terms

S12.1 Ethics provenance

IEEE requires that where ethical approval was obtained for research involving human subjects, the source and reference appear in the article itself. This work collected no new data; every corpus was released for public reuse by its original investigators. The nine corpora do not report approval uniformly, and Table S13 states for each what its own source publication or distribution record says. Where a source does not report an approval, we say so rather than implying one exists. We have not inferred, and do not report, any approving body or protocol number that a source does not itself state.

Table S13: Ethics provenance per corpus, taken only from what each source states. “Not stated in source” means the source publication or dataset record available to us does not report an approval; it is not a claim that none was obtained.

Corpus	Use here	Ethics provenance as stated by the source
ds002721	EEG reliability; individual-level analysis	Distributed via OpenNeuro with participant consent for public release; approval as reported in the dataset publication [7]. The dataset’s <code>dataset_description.json</code> records CC0 while its README records CC BY 4.0; we follow the more restrictive terms and attribute.
ds003690	Measured design requirement	Distributed via OpenNeuro under CC0 with approval as reported in the originating publications [8, 9].
PMEmo	Within-corpus ceiling; electrodermal analysis; per-person calibration	Human participants provided ratings and electrodermal recordings; approval as reported in the corpus publication [10].
BIRAFFE2	Withdrawn methodological case (Section S2)	Approval as reported in the corpus publication [1].
ARAUS	Per-person calibration	Human listeners rated soundscapes in a laboratory; approval as reported in the corpus publication [11].
DEAM	Transfer; per-person calibration	Annotation by crowdsourced raters; no approval statement located in the source [12].
Soundtracks	Transfer; the tonality result	Ratings collected from human listeners; no approval statement located in the source [13].
Emo-Soundscapes	Transfer	Ratings collected by pairwise comparison from human listeners; no approval statement located in the source [14].
ESC-50	Category control	Environmental sound recordings; no human-subject data and no approval applicable [15].

S12.2 Accession, access date and licence

Table S14 lists, per corpus, the accession identifier we retrieved it under, the date of retrieval, and the terms it is distributed on. The main text points here rather than carrying the list, because the licence terms differ per corpus in ways a single sentence cannot summarise – in particular, whether *derived* files may be redistributed is a different question from whether the audio may be, and the two answers diverge for four of the nine.

Derived tables underlying every figure panel are provided as supplementary source data.

Table S14: Data availability. All analyses use publicly available data; no new data were generated.

Corpus	Accession, access date and terms
ds002721	OpenNeuro ds002721 , retrieved 2026-07-27, https://openneuro.org/datasets/ds002721 . Licence discrepancy noted in Table S13; treated as CC BY 4.0 [7]. Its stimuli are excerpts from the Soundtracks set, so it and our fourth music corpus are not independent material.
ds003690	OpenNeuro ds003690 , retrieved 2026-07-30, https://openneuro.org/datasets/ds003690 . CC0 [16]; distributors request attribution to [8] for EEG and pupil data and [9] for the electrocardiogram.
Soundtracks Set 1	Open Science Framework p6vkg , retrieved 2026-07-27, https://osf.io/p6vkg/ . The deposit is CC BY 4.0 [13]; the excerpts are taken from commercially released film scores, so the deposit’s licence does not convey rights in the underlying recordings and we treat the audio as usable for research only.
DEAM	Distributed by the University of Geneva; audio licences vary per track [12].
PMemo	Annotations and precomputed features are redistributable; the audio consists of commercial excerpts and is not [10].
Emo-Soundscapes	Assembled from Creative Commons material on Freesound; per-item licences are given in the dataset metadata [14].
ESC-50	CC BY-NC 3.0 [15].
ARAUS	Distributed by Nanyang Technological University via DR-NTU, retrieved 2026-08-03. CC BY-NC 4.0 [11]. Used for the per-person calibration analysis only; the non-commercial term means the material itself cannot be reused in an application, whereas the calibration requirement estimated from it can.
BIRAFFE2	CC BY 4.0 [1]; used only for the withdrawn methodological case in Section S2.

Because these are small tabular files rather than a primary dataset, they accompany the manuscript rather than being deposited in a community repository.

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